

Basic Vaccination Guide (BVG)

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Thanks God for blessing this work.

This is a simple review summary of vaccines in Egypt.

Compulsory vaccines have been presented in short videos.
The duration of each presentation is about 3 minutes to be easily absorbed by audience.

Special thanks to Dr.Amir Lewiz for his great effort to upload each video one by one and make them easily accessible on YouTube through the hashtag #Vaccine_in3minutes.

Extra vaccines have been presented in online recorded lecture of EUP Clinic Course and a recorded lecture about Pitfalls in Vaccination practice has been presented in The First EUP Conference.

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Introduction

Minimum intervals – how to use them ;

- When vaccine doses are given too close together , the body may have insufficient time to mount an immune response , and the vaccines efficacy cannot be assured.
- It's preferable to maintain the longer "recommended" interval whenever possible , with a few exceptions.
- You should intentionally schedule patients at minimum intervals only in certain circumstances:

- When catching up a child who starts his/ her immunizations late or more than one month behind schedule .
- When a patient is scheduled to travel internationally.
- When the patient is unlikely to return for recommended visits.

4 – day grace period ?

The ACIP has established one small exception to minimum intervals – " the 4-day grace period – that allows vaccine doses administered up to four days before the minimum interval or age to be counted as valid. However , the 4-day grace period doesn't apply in all situations. It shouldn't be used when scheduling future vaccination visits . It also shouldn't be applied to the 28-day interval recommended when live parenteral vaccines are not administered at the same visit or when giving rabies vaccine. The grace period should be used primarily when reviewing historical vaccination records.

N.B : In cases where the vaccine doses given less than 28 days apart are two doses of the same live vaccine in a series (e.g., 2 doses of MMR vaccine), not different vaccines , you don't need to repeat the second dose if it was inadvertently administered within the 4-day "grace period" before day 28. If given more than 4 days earlier than day 28, the second dose should be repeated after the recommended minimum interval from the invalid dose. (immunize.org)

Maximum intervals – they don't exist

- Let's clear up one popular misconception ; there is no such a thing as a "maximum interval''. With very limited exceptions (e.g. oral typhoid vaccine in certain circumstances), you do NOT need to restart a vaccine series because an interval is longer than recommended.

- Doses given even years later than recommended are still valid because the body has " immunologic memory". The real problem with longer than recommended intervals is not the validity of the doses or their immunologic effect . It's that, until the series is complete, the person may remain susceptible to the associated vaccine- preventable disease.

GENERAL RULES

قواعد عامة في التطعيمات

- **Inactivated vaccines don't interfere with the immune response to other inactivated vaccines or to live vaccines. So any inactivated vaccine can be administered either concurrently or at any time before or after a different inactivated vaccine or live vaccine.**

- **Two exceptions , are :**

1. **A 4- week interval between PCV13 and MenACWY-D in a person with anatomic asplenia.**
2. **The separation of doses between PCV13 and PPSV23 :**

- **6 – 12 months recommended for non-high risk,(8 week minimum) if PCV13 is given first.**
- **8 weeks in children 6- 18 years , and 1 year minimum in adult 19 years and older if PPSV23 is given first.**

GENERAL RULES

قواعد عامة في التطعيمات

- The immune response to one live-virus vaccine might be impaired if administered within 28 days (i.e., 4 weeks) of another live- virus vaccine.
- Two or more injectable administered live vaccines not administered on the same day should be separated by at least 4 weeks , to minimize the potential risk for interference.
- If 2 such vaccines are separated by < 4 weeks , the second vaccine administered should not be counted and the dose should be repeated at least 4 weeks later.
- On the day a live injectable vaccine will be administered , providers should ensure that no live injectable vaccine was given in the previous 28 days.
- The oral vaccines Ty21a typhoid vaccine and rotavirus can be administered simultaneously with or at any interval before or after other live vaccines (injectable or intranasal) if indicated.

Antigen Combination	Recommended minimum interval between doses
Two or more inactivated	May be administered simultaneously or at any interval between doses
Inactivated and live	May be administered simultaneously or at any interval between doses
Two or more live injectable	28 days minimum interval , if not administered simultaneously

Vaccination Schedule

جدول التطعيمات الإجبارية

Compulsory vaccination schedule

Dose No		Vaccine		Age
Zero dose	جرعة صفرية	Hepatitis B	الكبدي ب	1st day اليوم الاول
Once Zero dose	جرعة واحدة جرعة صفرية	BCG OPV	درن شلل اطفال نقط	1st month الشهر الاول
1st	الاولى	OPV Hexavalent or Pentavalent + IPV One dose includes : -DTP -Hib -Hepatitis B -IPV	 ثلاثي انفلونزا بكتيرية الكبدي ب شلل اطفال حقن	2 months شهرين
2nd	الثانية	Hexavalent or Pentavalent + IPV One dose includes : -DTP -Hib -Hepatitis B -IPV	 ثلاثي انفلونزا بكتيرية الكبدي ب شلل اطفال حقن	4 months 4 شهور
3rd	الثالثة	Hexavalent or Pentavalent + IPV One dose includes : -DTP -Hib -Hepatitis B -IPV	 ثلاثي انفلونزا بكتيرية الكبدي ب شلل اطفال حقن	6 months 6 شهور
4th	الرابعة	OPV	شلل اطفال نقط	9 months 9 شهور
1st Booster	الاولى منشطة	MMR OPV	ام ام ار شلل اطفال نقط	12 months 12 شهر
Booster 2nd Booster	منشطة الثانية منشطة	DTP MMR OPV	الثلاثي ام ام ار شلل اطفال نقط	18 months 18 شهر

جدول التطعيمات الإضافية الضرورية

Additional recommended vaccination schedule

Dose No		Vaccine		Age
1st	الاولى	PCV Rota (e.g, Rotarix)	مكورات رئوية روتا	2 months شهرين
2nd	الثانية	PCV Rota (e.g, Rotarix)	مكورات رئوية روتا	4 months 4 شهور
3rd	الثالثة	PCV	مكورات رئوية	6 months 6 شهور
1st	الاولى	Meningococcal Conjugate Quadrivalent (e.g,Menactra) حمى شوكية رباعي مقترن		9 months 9 شهور
1st	الاولى	Chickenpox Hepatitis A	الجديري المائي الكبدي أ	12 months 12 شهر
2nd	الثانية	Meningococcal Conjugate Quadrivalent (e.g,Menactra) حمى شوكية رباعي مقترن		
Booster	منشطة	PCV	مكورات رئوية	12-15 months 15-12 شهر
Booster	منشطة	Hib	انفلونزا بكتيرية	18-24 months 24-18 شهر
2ND	الثانية	Hepatitis A Chickenpox	الكبدي أ الجديري المائي	
1st	الاولى	HPV	سرطان عنق الرحم	9 years 9 سنوات

Compulsory Vaccines

Compulsory vaccines schedule in Egypt (updated 22 / 2 / 2021)

Age	Dose	Vaccine	Disease	Dose	Mode of administration.
At birth	Birth dose	Hepatitis B	Hepatitis B	0.5 ml	IM anterolateral aspect of RT thigh
At birth	Birth	Sabin (OPV) BCG	Poliomyelitis TB	2 Drops 0.5 ml	Oral S.C UPPER LT ARM
2 MONTHS	FIRST	Sabin (OPV) Pentavalent Salk (IPV)	Poliomyelitis DTP , HBV , HIB Poliomyelitis	2 Drops 0.5 ml 0.5 ml	Oral IM RT thigh (ant.lat) IM LT thigh (ant.lat)
4 MONTHS	SECOND	Sabin (OPV) Pentavalent Salk (IPV)	Poliomyelitis DTP , HBV , HIB Poliomyelitis	2 Drops 0.5 ml 0.5 ml	Oral IM RT thigh (ant.lat) IM LT thigh (ant.lat)
6 MONTHS	THIRED	Sabin (OPV) Pentavalent Salk (IPV)	Poliomyelitis DTP , HBV , HIB Poliomyelitis	2 Drops 0.5 ml 0.5 ml	Oral IM RT thigh (ant.lat) IM LT thigh (ant.lat)
9 MONTHS	FOURTH	Sabin (OPV)	Poliomyelitis	2 Drops	Oral
12 MONTHS	FIFTH	Sabin (OPV) MMR	Poliomyelitis MMR	2 Drops 0.5 ml	Oral S.C UPPER RT ARM
18 MONTHS	BOOSTER	Sabin (OPV) MMR DTP	Poliomyelitis MMR DTP	2 Drops 0.5 ml 0.5 ml	Oral S.C UPPER RT ARM IM LT thigh (ant.lat)

Hepatitis B Virus Vaccine



What are the recommended doses of hepatitis B vaccine in the compulsory vaccine schedule in Egypt ?

- **Birth dose : Newborns should receive their first dose of vaccine within the first 24 hours of birth. Newborns who for a medical or other reason did not get the vaccine at birth should get as soon as possible . The benefits of the birth dose can still be achieved if the hepatitis B vaccine is given within the first seven days of life. If the vaccine is not given within this time frame, the baby may start the course of hepatitis B vaccines (in combination with other vaccines) beginning at 2 , 4 , 6 months of age.**
- **Other recommended 3 doses are given at 2, 4 , 6 months of age.**
- **The schedule for hepatitis B is flexible, but minimal intervals and minimum ages need to be observed :there should be at least 4 weeks between dose 1 and 2,and at least 8 weeks between doses 2 and 3.The minimum interval for the overall series from dose 1 to final dose is 4 months (16 weeks).**

Is it harmful to have an extra dose of hepatitis B vaccine ?

- No , getting extra doses of hepatitis B vaccine is not harmful.
- Hepatitis B vaccine does not contain any live virus. It is a recombinant DNA vaccine and can not cause hepatitis B.

Special situations

What is the current recommendation for a baby born to a mother with hepatitis B ?

- Infants born to mothers who are hepatitis B surface antigen-positive are recommended to receive both hepatitis B vaccine and HBIG.Both should be given on the day of birth, at the same time but in separate thighs .
- Infants should receive 3 subsequent doses of hepatitis B vaccine at 2 , 4 , 6 months of age , so that they receive a total of 4 doses of hepatitis B vaccine.
- The dose of HBIG (hepatitis B immunoglobulin) is 100 IU given by intramuscular injection

What is hepatitis B immunization management of preterm infants weighing < 2000 g ,by maternal hepatitis B surface antigen positive status ?

- Give hepatitis B vaccine and HBIG Within 12 hours of birth .
- Do not count the birth dose as part of the vaccine series.
- Give 3 additional hepatitis B vaccine doses at 2 , 4 , 6 months of age.

What is hepatitis B immunization management of preterm infants weighing < 2000 g ,by maternal hepatitis B surface antigen negative or unknown status ?

- If negative ; Delay first dose of hepatitis B until age 1 month or hospital discharge and complete the hepatitis B vaccine series at ages 2 , 4 , 6 months.
- If unknown ; give HBIG + hepatitis B vaccine within 12 hrs of birth.

- Test the mother for HBsAg.
- Do not count the birth dose as part of the vaccine series.
- Give 3 additional hepatitis B vaccine doses at ages 2 , 4 , 6 months.

BCG Vaccine

What is the recommended dose in the compulsory schedule in Egypt ?

- **One dose** soon after birth.

What are the guidelines for tuberculin skin test and BCG Vaccination ?

According to Eastern Regional Health Authority (ERHA) Guidelines for BCG immunization :

- **if under 3 months of age and no contact with TB , give BCG without tuberculin skin test.**
- **Children from 3 months of age and older , A tuberculin skin test should be carried out before BCG immunization. If negative : give BCG.**

Criteria for TB testing : (McMaster University , Canada)

- 1 or more months in country with TB incidence ≥30/100,000 population with high –risk contact, particularly direct patient contact in a hospital or indoor setting , but possibly including work in prisons , homeless shelters , refugee camps or inner-city slums.
- 3 or more months in country with TB incidence > 400/100,000 population
- 6 or more months in country with TB incidence 200-399/100,000 population
- 12 or more months in country with TB INCIDENCE 100-199/100,000 population
- Notified by Occupational Health or Public Health Services about significant exposure to an individual with active TB since the last negative TST/IGRA.

The BCG vaccine isn't given as part of the routine NHS vaccination schedule

- It's given on the NHS only when a child or adult is thought to have an increased risk of coming into contact with TB.
- When preparing for travel abroad , the BCG vaccine is recommended for any unvaccinated people under 16 who'll be living or working with friends , family or local people for more than 3 months in a country where TB is common or the risk of multi-drug resistant TB is high.
- TST should not be done routinely before BCG vaccination in children younger than 6 years unless they have a history of residence or prolonged stay (more than 1 month) in a country with a high incidence of tuberculosis.
- Areas of the world where the risk of TB is high enough to recommend BCG vaccination for previously unvaccinated travellers include : Bangladish , Pakistan , India , Africa , parts of south and southeast Asia , parts of South and Central America , parts of the Middle East.

UK Health Security Agency

Routine childhood immunisations

From September 2023

Age due	Diseases protected against	Vaccine given and trade name		Usual site[1]
Eight weeks old	Diphtheria, tetanus, pertussis (whooping cough), polio, Haemophilus influenzae type b (Hib) and hepatitis B	DTaP/IPV/Hib/HepB	Infanrix hexa or Vaxelis	Thigh
	Meningococcal group B (MenB)	MenB	Bexsero	Left thigh
	Rotavirus gastroenteritis	Rotavirus	Rotarix[2]	By mouth
Twelve weeks old	Diphtheria, tetanus, pertussis, polio, Hib and hepatitis B	DTaP/IPV/Hib/HepB	Infanrix hexa or Vaxelis	Thigh
	Pneumococcal (13 serotypes)	PCV	Prevenar 13	Thigh
	Rotavirus	Rotavirus	Rotarix[2]	By mouth
Sixteen weeks old	Diphtheria, tetanus, pertussis, polio, Hib and hepatitis B	DTaP/IPV/Hib/HepB	Infanrix hexa or Vaxelis	Thigh
	MenB	MenB	Bexsero	Left thigh
One year old (on or after the child's first birthday)	Hib and MenC	Hib/MenC	Menitorix	Upper arm/thigh
	Pneumococcal	PCV booster	Prevenar 13	Upper arm/thigh
	Measles, mumps and rubella (German measles)	MMR	MMRvaxPro[3] or Priorix	Upper arm/thigh
	MenB	MenB booster	Bexsero	Left thigh
Eligible paediatric age group[4]	Influenza (each year from September)	Live attenuated influenza vaccine LAIV	Fluenz Tetra[3,5]	Both nostrils
Three years four months old or soon after	Diphtheria, tetanus, pertussis and polio	dTaP/IPV	Boostrix-IPV	Upper arm
	Measles, mumps and rubella	MMR (check first dose given)	MMRvaxPro[3] or Priorix	Upper arm
Boys and girls aged twelve to thirteen years	Cancers and genital warts caused by specific human papillomavirus (HPV) types	HPV[6]	Gardasil 9	Upper arm
Fourteen years old (school Year 9)	Tetanus, diphtheria and polio	Td/IPV (check MMR status)	Revaxis	Upper arm
	Meningococcal groups A, C, W and Y	MenACWY	Nimenrix	Upper arm

1. Intramuscular injection into deltoid muscle in upper arm or anterolateral aspect of the thigh.
2. Rotavirus vaccine should only be given after checking for SCID screening result.
3. Contains porcine gelatine.
4. See annual flu letter at: www.gov.uk/government/collections/annual-flu-programme
5. If LAIV (live attenuated influenza vaccine) is contraindicated or otherwise unsuitable use inactivated flu vaccine (check Green Book Chapter 19 for details).
6. See Green Book chapter 18a for immunising immunocompromised young people who will need 3 doses.

Selective immunisation programmes

Target group	Age and schedule	Disease	Vaccines required
Babies born to hepatitis B infected mothers	At birth, four weeks and 12 months old[1,2]	Hepatitis B	Hepatitis B (Engerix B/HBvaxPRO
Infants in areas of the country with TB incidence >= 40/100,000	Around 28 days old[4]	Tuberculosis	BCG
Infants with a parent or grandparent born in a high incidence country[3]	Around 28 days old[4]	Tuberculosis	BCG
Children in a clinical risk group	From 6 months to 17 years of age	Influenza	LAIV or inactivated flu vaccine if contraindicated to LAIV or under 2 years of age
Pregnant women	At any stage of pregnancy during flu season	Influenza	Inactivated flu vaccine
	From 16 weeks gestation	Pertussis	dTaP/IPV (Boostrix-IPV)

1. Take blood for HBsAg at 12 months to exclude infection.
2. In addition hexavalent vaccine (Infanrix hexa or Vaxelis) is given at 8, 12 and 16 weeks.
3. Where the annual incidence of TB is >= 40/100,000 – see www.gov.uk/government/publications/tuberculosis-tb-by-country-rates-per-100000-people
4. Check SCID screening outcome before giving BCG.

For vaccine supply information for the childhood programme please visit portal.immform.phe.gov.uk and check vaccine update for all other vaccine supply information.

immunisation | The safest way to protect children and adults

NHS

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Special situations

Infants with no scar should be revaccinated ?

- A scar will appear in most people vaccinated with BCG but not everyone.
- If the child didn't have this reaction it doesn't mean that they have not responded to it.
- Tuberculin skin test should be carried out. If negative : give a second BCG vaccination. This revaccination is given in the RIGHT arm. It's not necessary to inspect this scar. Advise not to have any vaccination in the right arm for three months after BCG. (Eastern Regional Health Authority (ERHA) Guidelines for BCG immunization)

What are the contraindications to BCG vaccination ? (According to Eastern Regional Health Authority (ERHA) Guidelines for BCG immunization)

1.Systemic corticosteroid therapy (other than as replacement) or other immunosuppressive treatment including radiotherapy.Children:Prednisolone 2 mg/kg daily for more than two weeks , or 1 mg /kg daily for more than one month. N.b : Children on lower daily doses of systemic corticosteroids for less than two weeks and those on alternate day regimens for longer periods ,may be given BCG. Inhaled steroids are not a contraindication.

2. Lymphoma or malignant neoplasms involving bone marrow or the lymphoreticular system , or gammaglobulin deficiency or abnormality.

3. Pyrexia > 38 .

4.Generalized eczema .

5.Generalized septic skin condition e.g infected dermatitis.

6.Positive tuberculin test.

7.HIV positivity. Infants born to women known to be HIV positive should have BCG deferred until 2 HIV PCR tests, the second of which is at or after six weeks of age , are negative.

8.Within 3 months of blood or plasma transfusion.

Bear in mind ;

- **According to Canadian immunization guide ; There is minimal or no interaction between blood products or Ig preparations , and :**
- **BCG vaccine**
- **Yellow fever vaccine**
- **Live intranasal vaccine (live attenuated influenza vaccine)**
- **Live oral vaccines (rotavirus , oral typhoid vaccines)**
- **inactivated vaccines**

- **According to Eastern Regional Health Authority (ERHA) Guidelines for BCG immunization In Maternity Hospitals :**
- **Jaundice is not a contraindication to BCG vaccination provided the baby is feeding well and in a post-natal ward.**
- **Babies receiving phototherapy should not be vaccinated.**

What's the best management of BCG abscess ?

*** DO NOT INCISE AND DRAIN.**

-Use warm water compresses over the injection site or suppurating lymph node(s) 4-5 times /day.

-Cover with dry (non –occlusive) dressing to keep clean. Swab for culture and sensitivity (C&S) ONLY if secondary infection is suspected.

-Antibiotic treatment , such as isoniazid (INH), is NOT routinely indicated.

- **Regional adenopathy in the absence of erythema or vesicle formation should be considered an expected reaction to the vaccine.**
- **With care , the BCG vaccination site should not cause any problems. After about 8 to 12 weeks , although sometimes longer , the scab should come off on its own and leave a small red lesion which will fade slowly over time and eventually leave a small scar. Occasionally healing can take up to 3 to 6 months. If the wound is not healed in excess of 6 months or you have a larger abscess at the site, another way of management should be considered.**
- **Surgical excision of the abscess with involved lymph node(s) is one of the choices for BCG vaccine-induced suppurative lymphadenitis , but special attention should be paid to controlling the surgical indications. Intraoperative meticulous manipulation and postoperative care are crucial to achieve a good outcome.**

Diphtheria, Tetanus , and pertussis (DTaP) Vaccine

What are the recommended doses in routine vaccination ?

- 5-dose series at age 2,4,6,15-18 months, 4-6 years.
  *Dose 4 may be administered as early as age 12 months if at least 6 months has elapsed since dose 3.
- Td vaccine is administered in schools in Egypt (grade 2 , grade 4)

What are the minimum intervals between doses in catch-up vaccination ?

- **Minimum intervals between doses for children (4 months through 6 years) whose vaccination has been delayed.**
- **Dose 1 to Dose 2 : 4 weeks**
- **Dose 2 to Dose 3 : 4 weeks**
- **Dose 3 to Dose 4 : 6 months**
- **Dose 4 to Dose 5 : 6 months**
- ***Dose 5 is not necessary if dose 4 was administered at age 4 years or older and at least 6 months after dose 3.**

Special situations

What is the guidance on the best approach to vaccination following inadvertent administration of Tdap instead of pediatric DTaP ?

- If Tdap is inadvertently administered instead of pediatric DTaP to a child aged less than 7 years as any one of the first three doses of the tetanus-diphtheria-pertussis vaccination series ,the Tdap dose should not be counted as valid ,and a replacement dose of pediatric DTaP should be administered.
- If the indvertant administration is discovered while the child is still in the office, the pediatric DTaP can be administered during the same visit.
- If the child has left the office , some experts suggest administering the replacement dose of pediatric DTaP within approximately 72 hours, or administering it 4 weeks later to optimize the child immune response.
- The remaining doses of the pediatric DTaP series should be administered on the routine schedule ,with at least a 4 week interval between the replacement dose and the next dose.
- If Tdap was inadvertently administered as the fourth or fifth dose in the tetanus-diphtheria-pertussis vaccination series to a child aged less 7 years ,the Tdap dose should be counted as valid and does not need to be repeated.

What is the current recommendation for wound management in children less than age 7 years ?

- Wound management in children less than age 7 years with history of 3 or more doses of tetanus-toxoid-containing vaccine :For all wounds except clean and minor wounds ,administer DTaP if more than 5 years since last dose of tetanus-toxoid-containing vaccine.
- Tetanus immunoglobulin (TIG) is recommended for tetanus treatment and prophylaxis against tetanus following injury in patients whose tetanus immunization is incomplete or uncertain.

What is the current recommendation for wound management in persons age 7 years or older ?

- Wound management in persons age 7 years or older with history of 3 or more doses of tetanus –toxoid –containing vaccine :For clean and minor wounds , administer Tdap or Td if more than 10 years since last dose of tetanus-toxoid-containing vaccine; for all other wounds (dirty or tetanus prone),administer Tdap or Td if more than 5 years since last dose of tetanus-toxoid-containing vaccine.
- Tdap is preferred for persons age 11 years or older who have not previously received Tdap or whose Tdap history is unknown .
- If a tetanus-toxoid-containing vaccine is indicated for a pregnant adolescent , use Tdap.

Poliovirus Vaccine



What are the recommended doses in routine vaccination?

- **In compulsory vaccine schedule in Egypt :**
- **OPV (Sabin):At birth , 2 , 4 , 6 ,9 ,12 , 18 months.**
- **IPV (Salk) : 2 , 4 , 6 months.**

What are the minimum intervals between doses in catch up vaccination?

- **Minimum intervals between doses of IPV :**
- **Dose 1 to dose 2 : 4 weeks**
- **Dose 2 to dose 3 : 4 weeks if current age is < 4 years, 6 months (as final dose) if current age is 4 years or older.**
- **Dose 3 to dose 4 :6 months (minimum age 4 years for final dose)**

Special situations

Is there any special consideration for IPV vaccination before the recommended age in compulsory schedule ?

- In the first 6 months of life , use minimum ages and intervals only for travel to a polio-endemic region or during an outbreak.
- Minimum age for Dose 1 : 6 weeks
- Minimum interval between doses : 4 weeks after the first dose , 4 weeks after the second dose
- * a fourth dose 6 or more months after the third dose
- If the accelerated schedule cannot be completed before leaving, the remaining doses should be given in the affected country, or upon returning home at the intervals recommended in the accelerated schedule .

Should vaccine be withheld for patients on steroids ?

- Steroid therapies that are short term (less than 2 weeks) are not considered contraindications to the use of live virus vaccines (e.g OPV , MMR , Varicella)
- Many clinicians consider a dose equivalent to either 2 mg/kg of body weight or a total of 20 mg per day of prednisone for 2 or more weeks as sufficiently immunosuppressive to raise concern about safety of vaccination with live viral vaccines. Providers should wait at least 1 month after discontinuation of therapy or reduction of dose before administering a live virus vaccine.

Is there any interference on immunogenicity in case of co-administration of the oral polio and rotavirus vaccines ?

- Co-administration of the oral polio and rotavirus vaccines may be accompanied by an interference on immunogenicity.
- A minimum interval of 2 weeks is recommended for the administration of rotavirus vaccination and oral polio vaccine.

Bear in mind ;

Early clinical trials suggested that oral poliovirus vaccine , when administered concomitantly with rotavirus vaccine ,reduces rotavirus seroconversion rates after the 1st rotavirus dose with modest or nonsignificant interference after completion of the full rotavirus vaccine course.

According to Canadian guidelines :

- Household contacts who are vaccinated with OPV should not have contact with immunocompromised people for **6 weeks** after receipt of OPV.

IDSA

Table 8. Safety of Administration of Live Vaccines to Contacts of Immunocompromised Persons

Live Vaccine	Shedding of Agent? (site)	Transmissibility from Vaccinated Immunocompetent Person?	Recommendation for Administering Vaccines (When Indicated) to Healthy Immunocompetent Contacts of Immunocompromised Patients
Influenza, live, attenuated nasal	Yes (nasal secretions)	Rare (from 1 vaccinated toddler)	Administer (strong, low); vaccinated persons to avoid close contact with persons with hematopoietic stem cell transplant or severe combined immune deficiency for 7 d (weak, very low)
Measles, mumps, and rubella	Measles: no Mumps: no Rubella: yes (nasopharynx, in low titer; breast milk)	No, except mother-to-infant transmission of rubella vaccine virus via breast milk	Administer (strong, moderate)
Polio, oral	Yes (stool)	Yes, with rare cases of vaccine-associated paralytic poliomyelitis	Do *not* administer (strong, high)
Rotavirus, oral	Yes (stool)	Yes, but no reported cases of symptomatic infection in contacts	Administer (strong, low)
Typhoid, oral	No	No	Administer (strong, low)
Varicella	Yes (skin lesions)	Rare, limited to vaccinees with skin lesions	Administer (strong, moderate); if skin lesions develop, avoid close contact with immunocompromised persons
Yellow fever	No, except possibly shed in breast milk	Yes (at least 3 cases of encephalitis in infants exposed to the vaccine via nursing)	Administer (strong, moderate) except to women who are nursing
Zoster	Yes (rarely recovered from injection site vesicles)	Not reported	Administer to those aged ≥60 y (strong, moderate); if skin lesions develop, avoid close contact with immunocompromised persons

Measles , Mumps and Rubella (MMR) Vaccine

What are recommended doses in routine vaccination ?

- CDC : 2-dose series at age 12-15 months , age 4-6 years.
- Compulsory vaccine schedule in Egypt : 2- dose series at age 12 months , age 18 months.

What is the minimum interval between doses in Catch-up vaccination ?

Unvaccinated children and adolescents :

2-dose series at least 4 weeks apart.

Special situations

What is the current recommendation of MMR vaccination for international travelers?

➡ **Before any international travel**

Infants 6 months through 11 months of age should receive one dose of MMR vaccine. Infants who get one dose of MMR vaccine before their first birthday should get two more doses (one dose at 12 through 15 months of age and another dose separated by at least 4 weeks)

Unvaccinated children age 12 months or older : should receive 2-dose series at least 4 weeks apart before departure.

Note : According to IDSA guidelines , a third dose of MMR is now recommended during mumps outbreaks for people at increased risk for mumps. Measles , although eliminated from the United States , continues to be endemic around the world.

Which vaccines should be considered for spacing after administration of immunoglobulin preparations or blood products ?

- **MMR and Varicella containing vaccine**

What are the recommended intervals between administration of antibody-containing products and measles-or varicella-containing vaccine,by product and indication for vaccination ?

Product / indication	Dose (IgG/Kg) and route	Recommended interval before measles- or live varicella –containing vaccine administration
Blood transfusion – RBCs washed	**10 ml/kg ,negligible IgG/Kg IV**	**None**
Blood transfusion – RBCs , adenine –saline added	**10 ml/kg (10 mg IgG /Kg) IV**	**3 months**
Blood transfusion –Packed RBCs(hematocrit 65%)	**10 ml/kg (60 mg IgG /Kg) IV**	**6 months**
Blood transfusion – Whole blood (hematocrit35% - 50%)	**10 ml/kg (80-100 mg IgG /Kg) IV**	**6 months**
Blood transfusion-Plasma/platelet products	**10 ml/kg (160 mg IgG /Kg) IV**	**7 months**
Botulinum Immune Globulin Intravenous (Human)	**1.0 ml/kg (50 mg IgG /Kg) IV**	**6 months**
Cytomegalovirus IGIV	**150 mg/kg maximum**	**6 months**
Hepatitis A IG – Contact prophylaxis	**0.1 ml/kg (16.5 mg IgG /Kg) IM**	**6 months**
Hepatitis A IG –International travel < 1 month stay	**0.1 ml/kg (16.5 mg IgG /Kg) IM**	**6 months**
Hepatitis A IG – International travel ≥ 1 month stay	**0.2 ml/kg (33 mg IgG /Kg) IM**	**6 months**
Hepatitis B IG	**0.06 ml/kg (10 mg IgG /Kg) IM**	**3 months**
IGIV-replacement therapy for immune deficiencies	**300-400 mg/kg IV**	**8 months**
IGIV- Immune thrombocytopenic purpura treatment	**400 mg/Kg IV**	**8 months**
IGIV- Postexposure varicella prophylaxis	**400 mg/Kg IV**	**8 months**
IGIV-PE measles prophylaxis for immune compromised contacts	**400 mg/Kg IV**	**8 months**
IGIV- Immune thrombocytopenic purpura treatment	**1000 mg/Kg IV**	**10 months**
IGIV-Kawasaki disease	**2 g/Kg IV**	**11 months**
Measles prophylaxis IG-Standard (i.e nonimmunocompromised) contact	**0.50 ml/kg (80 mg IgG/Kg) IM**	**6 months**
Monoclonal antibody to RSV F protein (e.g Synagis [Medimmune])	**15 mg/kg IM**	**None**
Rabies IG	**20 IU/Kg (22 mg IgG/Kg) IM**	**4 months**
Tetanus IG	250 **units (10 mg IgG/Kg) IM**	**3 months**
Varicella IG	**125 units /10 kg (60-200 mg IgG /Kg) IM ,maximum 625 units**	**5 months**

According to Canadian guidelines :

- For optimum immune response to MMR or Varicella containing vaccine, the vaccine should be administered at least 14 days prior to administration of an Ig preparation or blood product , or the vaccine administration delayed until the antibodies in the Ig preparation or blood product have degraded(refer to table) . if the interval between the administration of any of these vaccines and subsequent administration of an Ig preparation or blood product is less than 14 days , or if these vaccines are administered before the antibody has degraded , repeat the vaccine dose after the recommended interval.

Haemophilus Influenzae type b Vaccine



What are the recommended doses in the routine vaccination ?

➨ **In the compulsory vaccine schedule in Egypt > 3 dose series : at 2 , 4 , 6 months**

- **Booster dose at 18 months of age (HIBREX)**
- **CDC recommends Hib vaccination for all children younger than 5 years old.**
- **CDC does not recommend Hib vaccination for most people 5 years or older unless they :**

- **Have certain medical conditions and are unvaccinated (children with HIV infection ,children with immunoglobulin deficiency , early component complement deficiency and children with asplenia)**
- **Receive a bone marrow transplant**

What are the minimum intervals between doses in catch-up vaccination ?

- Dose 1 at age 7-11 months : Administer dose 2 at least 4 weeks later and dose 3 at age 12-15 months or 8 weeks after dose 2 (whichever is later).
- Dose 1 at age 12-14 months : Administer dose 2 (final dose) at least 8 weeks after dose 1.
- Dose 1 before age 12 months and dose 2 before age 15 months :Administer dose 3 at least 8 weeks after dose 2.
- 1 dose administered at age 15 months or older : No further doses needed.
- Unvaccinated at age 15 month-59 months : Administer 1 dose.
- Previously unvaccinated children age 60 months or older who are not considered high risk : don't require catch-up vaccination.

Special situations

What are the recommended doses for children with Chemotherapy or Radiation treatment and children with Hematopoietic stem cell transplant (HSCT) ?

- **Chemotherapy or Radiation treatment :**

Age 12-59 months :

- Unvaccinated or only 1 dose before age 12 months : 2 doses, 8 weeks apart.

- 2 or more doses before age 12 months : 1 dose at least 8 weeks after previous dose. Doses administered within 14 days of starting therapy or during therapy should be repeated at least 3 months after therapy completion.

- **Hematopoietic stem cell transplant (HSCT):**

3 dose series 4 weeks apart starting 6 to 12 months after successful transplant , regardless of Hib vaccination history.

What are the recommended doses for children with asplenia ?

- Anatomic or Functional asplenia (including sickle cell disease) :

Age 12-59 months:

-Unvaccinated or only one dose before age 12 months : 2 doses 8 weeks apart.

-2 or more doses before 12 months : 1 dose at least 8 weeks after previous dose.

Unvaccinated *persons age 5 years or older : 1 dose.

- Elective splenectomy :

Unvaccinated *persons age 15 months or older : 1 dose (preferably at least14 days before procedure).

What are the recommended doses for children with HIV infection and children with immunoglobulin deficiency , early component complement deficiency ?

- **HIV infection :**

Age 12-59 months :

- Unvaccinated or only 1 dose before 12 months : 2 doses , 8 weeks apart.

- 2 or more doses before age 12 months: 1 dose at least 8 weeks after the previous dose.

Unvaccinated *persons age 5-18 years : - 1 dose.

- **immunoglobulin deficiency , early component complement deficiency :**

Age 12-59 months :

- Unvaccinated or only 1 dose before 12 months : 2 doses , 8 weeks apart.

- **2 or more doses before age 12 months: 1 dose at least 8 weeks after the previous dose.*Unvaccinated = Less than routine series (through age 14 months) or no doses (age 15 months or older).***

Extra Vaccines

- Rotavirus vaccine
- Pneumococcal vaccine
- Meningococcal vaccine
- Hepatitis A vaccine
- Varicella vaccine
- Influenza vaccine

اسم التطعيم	النوع	الجرعة	عمر الطفل
الروتا **Rotavirus vaccine**	عن طريق الفم	الاولى	شهرين
		الثانية	4 شهور
		التالتة	6 شهور
المكورات الرئوية **PCV**	حقن / عضل	الاولى	شهرين
		التانية	4 شهور
		التالتة	6 شهور
		المنشطة	12-15 شهر
الجديري المائي **Varicella vaccine**	حقن / تحت الجلد او عضل	الاولى	12- 15 شهر
		التانية	3 اشهر على الاقل بعد الجرعة الاولى
الكبدي الوبائي أ **HAV**	حقن / عضل	الاولى	12-23 شهر
		التانية	6-12 شهر بعد الجرعة الاولى
الحمى الشوكية **Meningococcal polysaccharide vaccine**	حقن / عضل	الاولى	سنتان
		التانية	4 -5 سنوات
الانفلونزا الموسمية Seasonal influenza	حقن/عضل		سنويا خلال فصل الخريف

Rotavirus vaccine

Rotavirus vaccination

(Minimum age : 6 weeks)

(Minimum interval between doses : 4 weeks)

➨ Routine vaccination

-Rotarix : 2-dose series at age 2 and 4 months

-Rotateq : 3-dose series at age 2, 4, 6 months

-If any dose in the series is either **Rotateq** or unknown,

Default to 3-dose series. (**they are interchangeable**)

Catch-up vaccination :

- Do not start the series on or after age 15 weeks , 0 days.
- The maximum age for the final dose is 8 months , 0 days.

According to the package inserts the maximum age for last dose of Rotateq is 32 weeks & the maximum age for Rotarix is 24 weeks. According to ACIP recommendations the maximum age for last dose of rotavirus vaccine is 8 months , 0 days. Eight months 0 days is older than age 24 weeks and may be older than age 32 weeks. Should I follow the package labels or the ACIP recommendations ?

- ACIP recommendations and package inserts do not always match.
- Occasionally, ACIP may use different data to formulate its recommendation, or try to add flexibility to its recommendation, which results in a recommendation different than the package insert.
- Published recommendations of national advisory groups (such as ACIP or AAP's committee on infectious diseases) should be considered equally as authoritative as those on the package insert. You should consider 8 months 0 days as the maximum age for a dose of rotavirus vaccine.

Precautions of Rotavirus vaccine administration :

- **Co-administration** of the oral polio and rotavirus vaccines may be accompanied by an interference on immunogenicity.
- A minimum interval of **2 weeks** is recommended for the administration of rotavirus vaccination and oral polio vaccine.
- Early clinical trials suggested that oral poliovirus vaccine , when **administered concomitantly** with rotavirus vaccine , reduces rotavirus seroconversion rates **after 1st rotavirus dose** with modest or **non-significant interference after completion of the full rotavirus vaccine course**.
- If an incomplete dose of an oral vaccine is administered for any reason (e.g , infant spits or regurgitates the vaccine) a replacement dose should not be administered. **" Pinkbook" , "Canadian Immunization Guide"**

Pneumococcal vaccine

Pneumococcal vaccination (minimum age : 6 weeks) : PCV 13

Prevenar 13 , Synflorix (10 serotypes)

Routine vaccination with PCV :

4-dose series at 2 , 4 , 6 , 12-15 months

Catch – up vaccination with PCV:

- **If the series has been started at 6 months of age or older and younger than one year : 3 dose series ,preferably 8 weeks apart (minimum 4 week interval) , then the last dose between the ages of 12 through 15 months (given at least 8 weeks after the second dose.**
- **If the series has been started at one year of age or older and younger than 2 years : 2 dose series (8 weeks apart).**
- **Healthy children age 24-59 months with any incomplete PCV series : 1 dose PCV (patients with immunocompromising or chronic medical conditions : 2 doses of PCV at least 8 weeks apart , give one dose of PPSV23 at least 8 weeks after last dose of PCV)**

- **N.B : PneuomoRecs VaxAdvisor application for pneumococcal vaccine Catch-up guidance (available for android and IOS users , approved by CDC).**

Can we consider PREVENAR 13 and SYNFLORIX interchangeable ?

- **Yes , The Theoretical serotype coverage of SYNFLORIX is similar to that of Prevenar.**
- **Pan American Health Organization (PAHO) ; Available evidence to date indicates significant impact of both PCV10 & PCV13 in the outcomes studied , with no evidence of the superiority of one vaccine over the other on pneumonia , IPD or meningitis hospitalization reduction in children under 5 years old.**

Countries or regions choose whether to use synflorix or prevnar 13 may decide to switch from one vaccine to the other

- **This can result in infants receiving a mixed vaccination regimen.**
- **What is new ? In a study the immunogenicity of three infant vaccination regimens had been assessed :**
  - **1) priming with two doses of Prevnar 13 and boosting with Synflorix;**
  - **2) priming one dose of Prevnar 13 followed by one dose of Synflorix and boosting with Synflorix;**
  - **3) priming and boosting with Synflorix.**
- **The study showed that : Switching from Prevnar 13 to Synflorix at any time during the vaccination regimen did not seem to affect safety. When switching from Prevnar 13 to Synflorix at the time of boosting , immunogenicity was mostly similar to that of the Synflorix-only regimen.**

Precautions of pneuomococcal vaccine administration :

- Giving **inactivated influenza** vaccine with **PCV13** increases **the risk for febrile convulsions in children** according to a vaccine safety Datalink study
- **Concomitant administration** is not preferred , They should be separated by **any interval (i.e,1 -2 weeks)** to **decrease the risk of febrile convulsions in children.**

Pneumococcal Polysaccharide Vaccine

Pneumovax 23

- **Indications :**
- **Adults 65 years of age and older**
- **Residents of Extended or Intermediate Care Facilities**
- **Individuals 2 years of age and older with HIV, Lymphoma , Hodgkins disease , Generalized malignancy , Chronic renal failure , Nephrotic syndrome , Receiving immunosuppressive therapy , organ or bone marrow transplant , Anatomic or functional asplenia, Sickle cell disease , Cystic fibrosis , Diabetes , Cochlear implant , Chronic neurological conditions that may impair clearance of oral secretions.**

- **Doses and schedule : 2 years of age and older : 1 dose given as 0.5 ml SC or IM , When both PCV and PPV23 are recommended , PCV should be given first , followed by at least 8 weeks later by PPV23. If PPV23 has already been administered , PCV should be administered one year later.**
- **Booster doses : A once only revaccination should be offered 5 years after the initial immunization to those who have Anatomic or functional asplenia, Sickle cell disease , Chronic kidney disease , Chronic liver disease including cirrhosis , chronic hepatitis B , and hepatitis C , Solid organ or islet cell transplant (candidate or recipient) , Immunosuppression related to disease [e.g , malignant neoplasm (including leukemia and lymphoma), HIV, Multiple myeloma] or therapy (e.g , high dose systemic steroids, or severe rheumatoid arthritis requiring immunosuppressive therapy) , Congenital immunodeficiency states.**

ONTARIO , 2024

Table 1. Pneumococcal vaccines currently available for use in children in Canada

Vaccine (Brand Name)	Manufacturer	Date of Authorization	Type of Vaccine	Pediatric Age Indication
PCV10* (Synflorix®)	GSK	December 11, 2008	Conjugate	6 weeks–5 years
PCV13 (Prevnar®13)	Pfizer	December 21, 2009	Conjugate	6 weeks–17 years
PCV15 (Vaxneuvance®)	Merck	July 8, 2022	Conjugate	6 weeks–17 years
PCV20 (Prevnar®20)	Pfizer	July 21, 2023	Conjugate	6 weeks–17 years
PPV23† (Pneumovax®23)	Merck	December 23, 1983	Polysaccharide	2 years–17 years

PCV=pneumococcal conjugate vaccine; PPV=pneumococcal polysaccharide vaccine

* Currently only used in Quebec

† Currently only children ≥2 years of age who have certain high-risk medical conditions are eligible to receive PPV23 in Ontario in addition to receiving an age-appropriate PCV13 series

In July 2022, Health Canada authorized PCV15 (Vaxneuvance®) in children 6 weeks to 17 years of age for the prevention of IPD caused by 15 *S. pneumoniae* serotypes. In July 2023, Health Canada authorized PCV20 (Prevnar®20) in children 6 weeks to 17 years of age for the prevention of IPD caused by 20 *S. pneumoniae* serotypes. Both PCV15 and PCV20 contain all 13 serotypes included in PCV13 (1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F), plus two additional serotypes (22F, 33F), with PCV20 containing five additional unique serotypes (8, 10A, 11A, 12F, 15B) (Table 2).

Varicella vaccine

Varicella vaccination

(minimum age : 12 months)

Routine vaccination

2 dose series : Varivax or Varilrix

They are interchangeable

Dosage		
Age	Children 12 months to 12 years	Adolescents and Adults 13 years and older
1st Dose	0.5 – ml dose administered subcutaneously (or IM)	0.5 -ml dose administered subcutaneously on elected date (or IM)
2nd Dose	0.5 – ml dose , a minimum interval of 3 months after the first dose (2 months " grace period ")	0.5 – ml dose to be administered , a minimum interval of 4 weeks after the first dose

Regardless of the duration of delay, interrupted schedules should be resumed as soon as possible without repeating previous doses.

Varicella Post Exposure prophylaxis:

- The Advisory Committee on Immunization Practices **(ACIP)** recommends that after being exposed to varicella or herps zoster , people **who do not have evidence of immunity** and are **eligible for vaccination should get varicella vaccine :**

- Ideally, the vaccine should be given within **3-5 days** after the person is exposed. This may prevent varicella or make it less severe.
- Even if it has been more than 5 days, the vaccine should still be offered. This will provide protection against varicella if a person is exposed again in the future.

Other types of post-exposure prophylaxis : UK Health Security Agency
- Antivirals (aciclovir or valacicolvir)

- **Oral aciclovir (or valaciclovir) is now the first choice of PEP for susceptible immunosuppressed individuals, all pregnant women at any age of pregnancy and infants at high risk. Oral acyclovir (or valaciclovir) should be given from day 7 to day 14 after exposure . The day of exposure is defined as the date of onset of the rash if the index is a household contact and date of first or only contact if the exposure is on multiple or single occasions respectively.**
- **If the patients presents after day 7 of exposure , a 7-day course of antivirals can be started up to day 14 after exposure , if necessary.**
- **In a study evaluating the comparative effectiveness of a 7-day course of aciclovir given either immediately after exposure or starting at day 7 after exposure to healthy children , the incidence and severity of varicella infection was significantly higher in those given aciclovir immediately after exposure (10 of 13 ; 77%) who received aciclovir immediately developed clinical varicella compared with 3 of 14 (21%) who started aciclovir at day 7.**
- **A 7-day post exposure prophylaxis course of aciclovir or valacicolvir is therefore recommended to start from day 7 after exposure.**
- **PEP following second exposure :**
  If there is a second or subsequent exposure to chickenpox further courses of antivirals can be initiating starting 7 days after the date of onset or exposure

Recommended doses of oral antivirals

	Oral aciclovir	Oral Valaciclovir
Infants over 4 weeks to children under 2 years age	10 mg/kg 4 times daily , day 7 to 14 after exposure	Not recommended
Children 2 to 17 years of age	10 mg /kg (up to a maximum of 800 mg) , 4 times daily, from days 7 to 14 after exposure	20mg/kg (up to a maximum 1000 mg) 3 times daily , from days 7 to 14 after exposure
Adults	800mg 4 times daily , from days 7 to 14 after exposure	1000mg 3 times daily , from days 7 to 14 after exposure

individuals on long term acyclovir or valaciclovir prophylaxis , for example , post-haematopoietic stem cell transplant , may require their dose of aciclovir to be temporarily increased to the dosage as given in Table.

Human varicella- zoster immunoglobulin (VZIG)

- For individuals who are unable to take oral antivirals (individuals with renal impairment or intestinal malabsorption , for example inflammatory bowel disease, and for susceptible neonates exposed within one week of delivery (in utero or post-delivery)VZIG should be considered
- If VZIG is considered , it's important to demonstrate that the patient will benefit from the blood product by demonstrating that they are sero-negative with VZV IgG antibody levels < 150 mIU / ml for immunosuppressed patients and < 100 mIU/ ml for pregnant women.
- For immunosuppressed patients only , if time does not permit quantitative testing , a qualitative test must be performed and shown to be negative or equivocal. Similarly for pregnant women who are unable to take antivirals due to renal impairment , intestinal malabsorption or hyperemesis , if time does not permit quantitative testing , a qualitative test must be performed or shown to be negative.

Dosage and timing of VZIG for prophylaxis

0 to 5 years	250 mg	By slow intramuscular injection
6 to 10 years	500 mg	
11 to 14 years	750 mg	
15 years and older	1000 mg	

When a large-volume injection such as VZIG is to be given , it should be administered deep into a large muscle mass. If more than 3 ml is to be given to young children and infants , or more than 5 ml to older children and adults , the immunoglobulin should be divided into smaller amounts and given into different sites . The upper outer quadrant of the buttock can be used to administer VZIG injection in neonates and infants.

VZIG should ideally be administered within 10 days (preferably 7 days for neonates and immunosuppressed contacts) of the day of exposure. The day of exposure is defined as the date of onset of the rash if the index is a household contact and date of first or only contact if the exposure is on multiple or single occasions respectively.

Further doses of VZIG following second exposure

➨ **Individuals who have previously received VZIG or IVIG as VZV post-exposure prophylaxis require a new risk assessment if a second exposure occurs :**

- **Within 3 weeks of administration of VZIG or IVIG , further PEP is not required**
- **between 3 and 6 weeks following administration of VZIG or IVIG , further PEP (dose of VZIG) should be administered without further testing**
- **More than 6 weeks following administration of VZIG or IVIG , retesting of a new sample is required**

Intravenous immunoglobulin (IVIG)

- Contacts with **bleeding disorders who cannot receive antivirals or be given an intramuscular injection** should given intravenous human normal immunoglobulin (IVIG) at a dose of 0.2 g per kg body weight (4 ml / kg for a 5 % solution) instead. This will produce serum VZV antibody levels equivalent to those achieved with VZIG.
- IVIG should ideally be administered within **10 days (preferably 7 days for neonates and immunosuppressed contacts)** , of the first contact, but can be given later if necessary.

Precautions of varicella vaccine administration :

- If 2 different live attenuated injectable vaccines are given < 28 days apart, the one given second should be repeated ≥ 28 days after the second or invalid dose. This would apply specifically to doses of **MMR** and **varicella**, if not administered on the same day.

Precautions of varicella vaccine administration :

- Because of the association between Reye syndrome , natural varicella infection and salicylates , the vaccine manufacturers advise against the use of salicylates for six weeks after VARICELLA VACCINE is given. There has been no reported association between the vaccine and Reye syndrome, but avoidance of salicylates is recommended as a precaution , and physicians need to weigh the theoretical risk of Reye syndrome from the vaccine against the known risk from varicella disease in children receiving long-term salicylate therapy. Children on low-dose aspirin following cardiac surgery would be more at risk of thrombosis from stopping their aspirin than from the theoretical risk of Reye's with VARICELLA VACCINE.
- People receiving long-term aspirin or salicylate therapy: (Australian Immunization Handbook)

- There is no need to avoid salicylates before or after receiving varicella or MMRV vaccine
- People taking long-term salicylate therapy (aspirin) can receive varicella –containing vaccine, if needed. The benefit is likely to outweigh any possible risk of Reye syndrome after vaccination.

What are the recommended intervals between administration of antibody-containing products and measles-or varicella-containing vaccine,by product and indication for vaccination ?

Product / indication	Dose (IgG/Kg) and route	Recommended interval before measles- or live varicella –containing vaccine administration
Blood transfusion – RBCs washed	**10 ml/kg ,negligible IgG/Kg IV**	**None**
Blood transfusion – RBCs , adenine –saline added	**10 ml/kg (10 mg IgG /Kg) IV**	**3 months**
Blood transfusion –Packed RBCs(hematocrit 65%)	**10 ml/kg (60 mg IgG /Kg) IV**	**6 months**
Blood transfusion – Whole blood (hematocrit35% - 50%)	**10 ml/kg (80-100 mg IgG /Kg) IV**	**6 months**
Blood transfusion-Plasma/platelet products	**10 ml/kg (160 mg IgG /Kg) IV**	**7 months**
Botulinum Immune Globulin Intravenous (Human)	**1.0 ml/kg (50 mg IgG /Kg) IV**	**6 months**
Cytomegalovirus IGIV	**150 mg/kg maximum**	**6 months**
Hepatitis A IG – Contact prophylaxis	**0.1 ml/kg (16.5 mg IgG /Kg) IM**	**6 months**
Hepatitis A IG –International travel < 1 month stay	**0.1 ml/kg (16.5 mg IgG /Kg) IM**	**6 months**
Hepatitis A IG – International travel ≥ 1 month stay	**0.2 ml/kg (33 mg IgG /Kg) IM**	**6 months**
Hepatitis B IG	**0.06 ml/kg (10 mg IgG /Kg) IM**	**3 months**
IGIV-replacement therapy for immune deficiencies	**300-400 mg/kg IV**	**8 months**
IGIV- Immune thrombocytopenic purpura treatment	**400 mg/Kg IV**	**8 months**
IGIV- Postexposure varicella prophylaxis	**400 mg/Kg IV**	**8 months**
IGIV-PE measles prophylaxis for immune compromised contacts	**400 mg/Kg IV**	**8 months**
IGIV- Immune thrombocytopenic purpura treatment	**1000 mg/Kg IV**	**10 months**
IGIV-Kawasaki disease	**2 g/Kg IV**	**11 months**
Measles prophylaxis IG-Standard (i.e nonimmunocompromised) contact	**0.50 ml/kg (80 mg IgG/Kg) IM**	**6 months**
Monoclonal antibody to RSV F protein (e.g Synagis [Medimmune])	**15 mg/kg IM**	**None**
Rabies IG	**20 IU/Kg (22 mg IgG/Kg) IM**	**4 months**
Tetanus IG	250 **units (10 mg IgG/Kg) IM**	**3 months**
Varicella IG	**125 units /10 kg (60-200 mg IgG /Kg) IM ,maximum 625 units**	**5 months**

According to Canadian guidelines :

- For optimum immune response to MMR or Varicella containing vaccine, the vaccine should be administered at least 14 days prior to administration of an Ig preparation or blood product , or the vaccine administration delayed until the antibodies in the Ig preparation or blood product have degraded(refer to table) . if the interval between the administration of any of these vaccines and subsequent administration of an Ig preparation or blood product is less than 14 days , or if these vaccines are administered before the antibody has degraded , repeat the vaccine dose after the recommended interval.

According to Canadian guidelines :

- There is **minimal or no interaction** between blood products or Ig preparations , and :
- Inactivated vaccines
- live oral vaccines (rotavirus , oral typhoid vaccines)
- live intranasal vaccine (live attenuated influenza vaccine)
- BCG vaccine
- Yellow fever vaccine

These vaccines may be given **concomitantly with , or at any time before or after** , an Ig preparation or blood product has been administered.

Hepatitis A vaccine

Hepatitis A vaccination :

(minimum age : 12 months for routine vaccination)

-The available products are Havrix 1440 , Havrix 720 Junior , Avaxim , and Avaxim Pediatric.

-These products are considered interchangeable at any scheduled dose for children or adults , using age-specific dosage for the particular product.

Routine vaccination :

-2 dose series given as 0.5 ml of Havrix 720 Junior IM (minimum interval : 6 months) at age 12- 23 months.

Catch up vaccination :

- **Unvaccinated people through age 18 years should complete a 2 dose series given as 0.5 ml of Havrix 720 Junior , IM (minimum interval : 6 months).**
- **Unvaccinated people who are 19 years of age and older should complete a 2 dose series given as 1 ml of Havrix 1440 , IM (minimum interval : 6 months).**
- **People who previously received 1 dose at age 12 months or older should receive dose 2 at least 6 months after dose 1.**
- **Resume the series no matter when the first dose was received , you don't need to restart a vaccine series because an interval is longer than recommended. Doses given even years later than recommended are still valid because the body has "immunologic memory"**

Special situation
(International travel)

- People travelling to or working in countries with high or intermediate endemic hepatitis A

- Infants age 6-11 months : 1 dose before departure ;

Revaccinate with 2 doses (separated by at least 6 months) between age 12- 23 months.

- Unvaccinated age 12 months or older : Administer dose 1 as soon as travel is considered.

Hepatitis A Post-Exposure Prophylaxis Guidance : (California Department of Public Health)

Postexposure prophylaxis (PEP)

All nonimmune people who are exposed to hepatitis A virus (HAV) and have not been previously infected or vaccinated should receive PEP within 14 days after the date of last exposure.

Definition of hepatitis A immunity:

People are considered immune if they have :

- received 2 doses of HAV vaccine; or

- a history of IgM or total anti-HAV positivity during or up to four months after clinically compatible illness ; or

-are igG anti-HAV positive

PEP recommendations

(California Department of Public Health)

- People ≥ 12 months of age , regardless of pre-existing medical conditions , SHOULD receive a dose of single-antigen HAV vaccine. In addition to vaccine , a dose of intramuscular (IM) immune globulin (IG) 0.1ml/kg > 40 years based on the provider's risk assessment.
- People ≥ 12 months of age who have chronic liver disease and/or immunocompromised SHOULD receive a dose of intramuscular immune globulin (IMIG) (0.1ml/kg) and a dose of HAV vaccine.
- Infants < 12 months of age and/or people who are allergic to vaccine component) SHOULD receive intramuscular immune globulin (IMIG) (0.1ml/kg) instead of HAV vaccine.
- People receiving both vaccine and IG for post-exposure prophylaxis may receive them simultaneously , or they may receive whichever product is available first and the second product when it is available, providing it is administered within the 14-day PEP window.
- Vaccine and IG should be administered at anatomically distant sites (such as different limbs).

Meningococcal vaccine

Different Types of Meningococcal Vaccines :

Mencevax ACWY:

- Meningococcal polysaccharide vaccine
- Neisseria meningitides serotypes A,C,W-135 and Y
- Not recommended in children < 2 years
- Repeated every 2-3 years in areas of high risk

N.B : Administered in schools in Egypt every 2-3 years

➡ Menactra (Serogroup : ACWY-DT)

- Meningococcal polysaccharide Diphtheria Toxoid conjugate vaccine
- Neisseria meningitides serotypes A,C,W-135 and Y
- 9-23 months : 2 doses , 3 months apart
- 2-55 years : single dose
- A single booster dose may be given to individuals 15-55 years at continued risk for meningococcal disease.

Nimenrix : Serogroup ACWY-TT

Age Group	Primary Immunisation	Booster
Infants from 6 weeks to less than 6 months of age	Two doses , each of 0.5 ml with the first dose given from 6 weeks of age , with an interval of 2 months between doses	At 12 months of age
Unvaccinated infants from 6 months to less than 12 months of age	One dose of 0.5 ml given from 6 months of age	At 12 months of age with a minimum interval of at least 2 months after the primary dose
Children from 12 months of age, adolescents and adults	One dose of 0.5 ml	Not routinely administered

Protection against meningococcal groups C , Y , and W , wanes more slowly than protection against meningococcal group A disease after immunization with Nimenrix. A booster dose of Nimenrix could be considered for children vaccinated as toddlers , five years after the previous dose if they continue to be at increased risk of meningococcal disease.

Can a different brand of quadrivalent (A,C,W_{135},Y)meningococcal conjugate vaccines (4vMenCV) be used to complete a vaccination course (i.e are the brands interchangeable), particularly if there are shortages in vaccine supply ?

- **It's preferable to use the same brand of 4vMenCV when giving subsequent doses , especially for completing a primary vaccination course for young children. However , a different brand may be substituted in cases where the initial brand used is unavailable or not known. (National Centre For Immunization Research & Surveillance, NCIRS Fact sheet , Australia)**

Special consideration for use of Meningococcal conjugate vaccine and prevenar 13 :

- Where both Prevenar 13 and Menactra are to be given , administration of Menactra must be at least 4 weeks before or after a dose of prevenar13.
- regarding use of Menactra for people with functional or anatomic asplenia or HIV :

- Before the age of 2 years ; children with functional or anatomic asplenia or HIV shouldn't receive Menactra. This timing avoids interference with immunologic response to the infant series of pneumococcal conjugate vaccine (PCV)
- People 2 years of age or older with functional or anatomic asplenia or HIV shouldn't receive Menactra at the same time as PCV , they should receive Menactra 4 weeks after completion of all PCV doses. If these vaccines are inadvertently administered on the same day , PCV13 should be repeated a minimum of 8 weeks later.

- Nimenrix may be co-administered with prevenar 13 (The Australian immunization Handbook).

Special considerations for concomitant use of Meningococcal conjugate vaccine and DTaP :

- Children can receive Menactra (ACWY-DT) or Nimenrix (ACWY-TT)before (at least 4 weeks before) or concomitantly with DTaP vaccines
- This timing avoids interference with the immunologic response to the meningococcal vaccine antigens that occurs when administering Menactra after DTaP.
- Even though ACIP recommends that Menactra should be given either before (at least 4 weeks before) , or at the same visit , or at least 6 months after DTaP, there is no evidence to support repeating the dose of Menactra if administered at a time shorter than that recommended interval.

N.B : The anti-diphtheria response was much higher after concomitant administration of Td with Menactra compared to the administration of Td followed by Menactra 28 days later.

(CDC , European Medicines Agency , Canadian immunization guide)

Special considerations for use of Meningococcal conjugate vaccine and polysaccharide meningococcal vaccine :

- Eligible individuals previously vaccinated with a polysaccharide meningococcal vaccine should be given meningococcal quadrivalent conjugate ; this should be offered at least 6 months after vaccination with polysaccharide meningococcal vaccine.

Influenza vaccine

Recommended doses of influenza vaccine by age group :

Influvac Tetra or Vaxigrip Tetra

Age	Dose	Number of doses needed in 1st year of influenza vaccination
6 months to < 9 years	0.5 ml	2 (given 4 weeks apart)
≥ 9years	0.5 ml	1
People of any age who have recently had haematopoietic stem cell transplant or solid organ transplant	0.5 ml	2 (given 4 weeks apart) in 1st year after transplant

Timing of vaccination

- For most people who need only one dose of influenza vaccine for the season , vaccination should ideally be offered during September or October. However , vaccination should continue throughout the season as long as influenza viruses are circulating.
- The Advisory Committee on Immunization Practices (The ACIP) recommends that all children 6 months old through 8 years old should get two doses of flu vaccine the first year that they are vaccinated the flu.
- Children 6 months through 8 years who require 2 doses should receive the first dose as soon as vaccine is available.
- Vaccination during July and August can be considered for children of any age who require only 1 dose.
- If the child (6 months old through 8 years old) had their first flu shot in a previous year but only got a single shot , then this year , they should get a flu shot and a booster shot. The second dose is given at least 28 days after the first dose.

Eligible groups for influenza vaccination:

- **People who are :**
- **Are 65 years of age and over**
- **are a child aged 2 to 12 years**
- **are pregnant**
- **all those aged 6-23 months and 13-64 years at increased risk of influenza-related complications such as:**

- **Those with chronic illness ,e.g. chronic heart disease (including acute coronary syndrome) chronic liver disease , chronic neurological disease (where the neurological condition compromises clearance of respiratory secretions), chronic renal failure , chronic respiratory disease (including chronic obstructive pulmonary disease , cystic fibrosis , moderate or severe asthma , and bronchopulmonary dysplasia), diabetes mellitus , or haemoglobinopathies**
- **Those with immunosuppression due to disease or treatment , including asplenia or hyposplenism , and all cancer patients.**
- **Those with any condition that can compromise respiratory function (e.g. spinal cord injury , seizure disorder , or other neuromuscular disorder) especially those attending special schools / day centres**
- **Children and adults with Down syndrome**
- **Children with moderate to severe neurodevelopmental disorders such as cerebral palsy and intellectual disability**
- **Those with morbid obesity (Body mass index > 40)**
- **Residents of nursing homes , old people's homes , and other long stay facilities where rapid spread is likely to follow introduction of infection**
- **Healthcare workers**
- **Household contacts of people with underlying chronic health condition or Down syndrome**
- **Out-of-home care givers for people who have an underlying chronic health condition or have Down syndrome.**
- **People in regular contact with pigs , poultry or waterfowl**

Interchangeability of influenza vaccines :

- Different age-appropriate brands **(Influvac Tetra or Vaxigrip Tetra)** are **interchangeable** for people who need 2 doses of influenza vaccine in a single season.

Special considerations for concomitant use of PCV13 , DTaP and Influenza vaccines :

- The findings showed that **PCV13** and **DTaP** both increase the risk of **febrile seizure** when given **at the same time as the influenza vaccine.**

Flu Vaccine and People with Egg Allergies :

- Studies have clearly demonstrated that egg-allergic people can receive influenza vaccines. All egg –allergic individuals may be vaccinated against influenza using any of the vaccines authorized for use.(Canadian Immunization Guide)
- Every 6 months and older with egg allergy should receive an annual flu vaccine (egg based or non-egg based) that is otherwise appropriate for the recipient's age and health status can be used. (CDC)

Human Papillomavirus vaccine

Human papillomavirus Vaccination (minimum age : 9 years)

➨ **Routine and catch-up vaccination :**

- **HPV vaccination routinely recommended at age 11-12 years (can start at age 9 years) and catch-up HPV vaccination recommended for all people through age 18 years if not adequately vaccinated.**
- **2- or 3-dose series depending on age at initial vaccination :**

- **Age 9-14 at initial vaccination : 2-dose series at 0, 6-12 months (minimum interval : 5 months ; repeat dose if administered too soon)**
- **Age 15 years or older at initial vaccination : 3-dose series at 0 , 1-2 months, 6 months (minimum intervals:dose1to dose 2: 4 weeks / dose 2 to 3 : 12 weeks / dose1 to dose 3 : 5 months; repeat dose if administered too soon)**

- **Interrupted schedules : If vaccination schedule is interrupted , the series does not need to be restarted.**
- **No additional dose recommended when any HPV vaccine series has been completed using the recommended dosing intervals**

➨ **Special situations :**

- **Immunocompromising conditions, including HIV infection :**

3-dose series , even for those who initiate vaccination at age 9 through 14 years.

- **History of sexual abuse or assault : Start at age 9 years.**

- **Two vaccines to prevent human papilloma (HPV) infection, the primary cause of cervical cancer , are now approved for use in many countries ; Cervarix and Gardasil.**
- **Some people Think of HPV immunization as only for girls but it's just as important for boys to get vaccinated.**
- **HPV vaccination protects boys from many kinds of cancer , most commonly from HPV-related cancers of the mouth and throat.**
- **The American Academy of Pediatrics recommends that HPV vaccination for healthy children can begin as early as age 9 years.**

HPV VACCINE SCHEDULE

GARDASIL 9-14 years of age		GARDASIL 15 -45 years of age	
Dose 1	At first visit	Dose 1	At first visit
		Dose 2	2 months after 1st dose
Dose 2	6-12 months after 1st dose	Dose 3	6 months after 1st dose

Health Education & Research Organization (HERO)

HPV VACCINE SCHEDULE

CERVARIX 9-14 years of age		CERVARIX 15 years and older	
Dose 1	At first visit	Dose 1	At first visit
		Dose 2	1 months after 1st dose
Dose 2	6 months after 1st dose	Dose 3	6 months after 1st dose

Health Education & Research Organization (HERO)

What is the difference between GARDASIL and CERVARIX ? Are they interchangeable ?

- **CERVARIX is a bivalent vaccine that protects against oncogenic(cancer causing) HPV types 16 and 18.**
- **GARDASIL vaccine is a quarivalent vaccine that protects against HPV types 16 and 18 as well as HPV types 6 and 11; the latter two are important causes of genital warts.**
- **Whenever possible , one brand of vaccine should be used to complete a vaccine series.**
- **If a person has received 1 or 2 HPV vaccine doses previously , and the brand of the previously received doses is not known , CERVARIX may be used to complete the series. Both vaccines provide protection against HPV types 16/18 and therefor clients are likely to achieve protective antibody levels against these HPV types. If less than 3 doses of GARDASIL are administered, counsel clients that protection against HPV types 6/11 (genital warts) cannot be assured.**

Covid-19 vaccine

Routine vaccination (minimum age : 6 months [Moderna & Pfizer BioNTech COVID--19 vaccines], 12 years [Novavax COVID-19])

- **Primary series :**

- **Age 6 months – 4 years : 2-dose series at 0,4-8 weeks (Moderna) or 3-dose series at 0, 3-8 , 11-16 weeks (Pfizer-BioNTech).**
- **Age 5-11 years : 2-dose series at 0,4-8 weeks (Moderna) or 2-dose series at 0, 3-8 weeks (Novavax, Pfizer – BioNTech).**
- **Age 12-18 years : 2-dose series at 0,4-8 weeks (Moderna) or 2-dose series at 0, 3-8 weeks (Novavax, Pfizer-BioNTech).**

Special situations :

People who are moderately or severely immunocompromised

- **Primary series**

- **Age 6 months – 4 years : 3-dose series at 0,4,8 weeks (Moderna) or 3-dose series at 0,3,11weeks (Pfizer-BioNTech).**
- **Age 5-11 years : 3-dose series at 0,4,8 weeks (Moderna) or 3-dose series at 0,3,7 weeks (Pfizer-BioNTech).**
- **Age 12-18 years : 3-dose series at 0,4,8 weeks (Moderna) or 2-dose series at 0,3 weeks (Novavax) or 3-dose series at 0,3,7 weeks (Pfizer-BioNTech).**

Are COVID-19 vaccines interchangeable ?

Yes and no

- **Children 6 months through 4 years old** should generally **stick** to the same COVID vaccine brand.
- **But people ages 5 years and older** can make the choice to switch up which vaccine they receive. The FDA and CDC are allowing **a "mix-and-match" approach for COVID booster shots in this older age group**. This means that regardless of the vaccine that someone received first , they can choose the vaccine that they want next. **For example** , if someone initially received **the original Pfizer COVID vaccine** , they can choose which vaccine they want for their next dose. They could decide to get **Pfizer again , or they could choose to get Moderna or Novavax.**
- If a person originally received the Pfizer vaccine, switching to the Moderna vaccine has been shown to result in a good immune response. The same goes for switching to Pfizer if a person originally got Moderna.

Summary of recent changes (last updated October 6 , 2023) :

- **The updated 2023-2024 formulation of Novavax COVID-19 Vaccine is recommended for people ages 12 years and older as follows :**
- **Initial vaccination : 2 doses of updated (2023-2024 formula) Novavax COVID-19 Vaccine.**
- **Previously vaccinated with any Original monovalent or bivalent COVID-19 (Moderna,Novavax, Pfizer-BioNTech, Janssen): 1 dose of updated (2023-2024 Formula) Novavax Vaccine.**
- **People who are moderately or severely immunocompromised may receive 1 or more additional updated (2023-2024 Formula) Novavax vaccine doses.**
- **People ages 12 years and older have the option of receiving either the updated (2023-2024 Formula) mRNA (Moderna,Pfizer-BioNTech) or updated (2023-2024 Formula) Novavax vaccine.**

Allergic reactions to vaccines and vaccine anaphylaxis

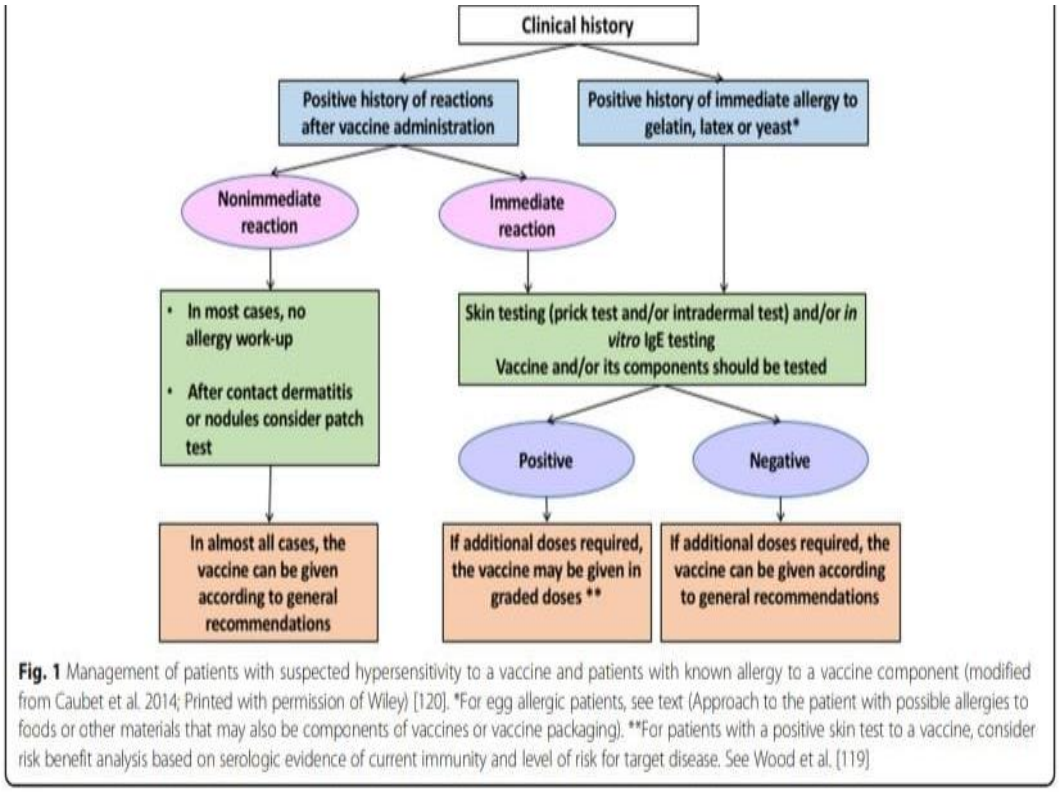


Fig. 1 Management of patients with suspected hypersensitivity to a vaccine and patients with known allergy to a vaccine component (modified from Caubet et al. 2014; Printed with permission of Wiley) [120]. *For egg allergic patients, see text (Approach to the patient with possible allergies to foods or other materials that may also be components of vaccines or vaccine packaging). **For patients with a positive skin test to a vaccine, consider risk benefit analysis based on serologic evidence of current immunity and level of risk for target disease. See Wood et al. [119]

International Consensus Guidelines (ICON) , Allergic reactions to vaccines.

World Immunization Journal

Resuscitation Council UK

GUIDELINES 2021

Anaphylaxis

Anaphylaxis?

A = Airway **B** = Breathing **C** = Circulation **D** = Disability **E** = Exposure

Diagnosis – look for:

- Sudden onset of Airway and/or Breathing and/or Circulation problems[1]
- And usually skin changes (e.g. itchy rash)

Call for HELP

Call resuscitation team or ambulance

- Remove trigger if possible (e.g. stop any infusion)
- Lie patient flat (with or without legs elevated)
  - A sitting position may make breathing easier
  - If pregnant, lie on left side



Give intramuscular (IM) adrenaline[2]

Inject at **anterolateral aspect** – middle third of the thigh



- Establish airway
- Give high flow oxygen
- Apply monitoring: pulse oximetry, ECG, blood pressure

If no response:

- Repeat IM adrenaline after 5 minutes
- IV fluid bolus[3]

If no improvement in Breathing or Circulation problems[1] despite TWO doses of IM adrenaline:

- Confirm resuscitation team or ambulance has been called
- Follow REFRACTORY ANAPHYLAXIS ALGORITHM

1. Life-threatening problems

Airway
Hoarse voice, stridor

Breathing
↑work of breathing, wheeze, fatigue, cyanosis, SpO_2 <94%

Circulation
Low blood pressure, signs of shock, confusion, reduced consciousness

2. Intramuscular (IM) adrenaline
Use adrenaline at 1 mg/mL (1:1000) concentration

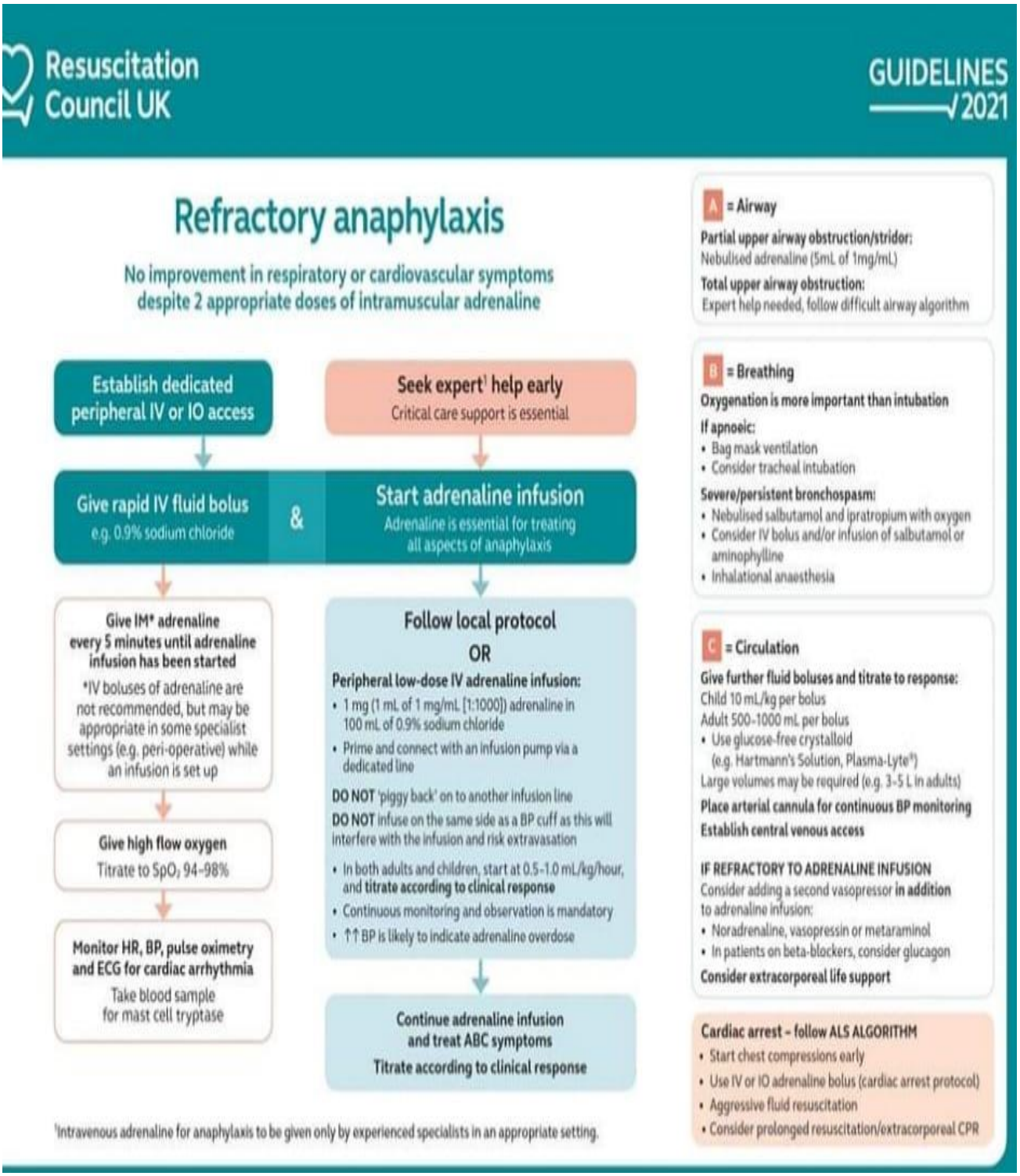
Adult and child >12 years:	500 micrograms IM (0.5 mL)
Child 6–12 years:	300 micrograms IM (0.3 mL)
Child 6 months to 6 years:	150 micrograms IM (0.15 mL)
Child <6 months:	100–150 micrograms IM (0.1–0.15 mL)

The above doses are for IM injection **only**.
Intravenous adrenaline for anaphylaxis to be given **only by experienced specialists** in an appropriate setting.

3. IV fluid challenge
Use crystalloid

Adults: 500–1000 mL
Children: 10 mL/kg

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Resuscitation Council UK
GUIDELINES 2021
Refractory anaphylaxis
No improvement in respiratory or cardiovascular symptoms despite 2 appropriate doses of intramuscular adrenaline
Establish dedicated peripheral IV or IO access
Seek expert[1] help early
Critical care support is essential
Give rapid IV fluid bolus
e.g. 0.9% sodium chloride
&
Start adrenaline infusion
Adrenaline is essential for treating all aspects of anaphylaxis
Give IM* adrenaline every 5 minutes until adrenaline infusion has been started
*IV boluses of adrenaline are not recommended, but may be appropriate in some specialist settings (e.g. peri-operative) while an infusion is set up
Give high flow oxygen
Titrate to SpO2 94–98%
Monitor HR, BP, pulse oximetry and ECG for cardiac arrhythmia
Take blood sample for mast cell tryptase
Follow local protocol
OR
Peripheral low-dose IV adrenaline infusion:
• 1 mg (1 mL of 1 mg/mL [1:1000]) adrenaline in 100 mL of 0.9% sodium chloride
• Prime and connect with an infusion pump via a dedicated line
DO NOT 'piggy back' on to another infusion line
DO NOT infuse on the same side as a BP cuff as this will interfere with the infusion and risk extravasation
• In both adults and children, start at 0.5–1.0 mL/kg/hour, and titrate according to clinical response
• Continuous monitoring and observation is mandatory
• ↑↑ BP is likely to indicate adrenaline overdose
Continue adrenaline infusion and treat ABC symptoms
Titrate according to clinical response
[1]Intravenous adrenaline for anaphylaxis to be given only by experienced specialists in an appropriate setting.
A = Airway
Partial upper airway obstruction/stridor:
Nebulised adrenaline (5mL of 1mg/mL)
Total upper airway obstruction:
Expert help needed, follow difficult airway algorithm
B = Breathing
Oxygenation is more important than intubation
If apnoeic:
• Bag mask ventilation
• Consider tracheal intubation
Severe/persistent bronchospasm:
• Nebulised salbutamol and ipratropium with oxygen
• Consider IV bolus and/or infusion of salbutamol or aminophylline
• Inhalational anaesthesia
C = Circulation
Give further fluid boluses and titrate to response:
Child 10 mL/kg per bolus
Adult 500–1000 mL per bolus
• Use glucose-free crystalloid (e.g. Hartmann's Solution, Plasma-Lyte®)
Large volumes may be required (e.g. 3–5 L in adults)
Place arterial cannula for continuous BP monitoring
Establish central venous access
IF REFRACTORY TO ADRENALINE INFUSION
Consider adding a second vasopressor in addition to adrenaline infusion:
• Noradrenaline, vasopressin or metaraminol
• In patients on beta-blockers, consider glucagon
Consider extracorporeal life support
Cardiac arrest – follow ALS ALGORITHM
• Start chest compressions early
• Use IV or IO adrenaline bolus (cardiac arrest protocol)
• Aggressive fluid resuscitation
• Consider prolonged resuscitation/extracorporeal CPR

The recommended protocol for further doses in case of allergic reactions to vaccines with positive skin test :

- **According to international consensus guidelines of allergic reactions to vaccines (ICON) , World immunization journal.**

Graded dosing protocol for remaining shots of vaccine. And a stand by treatment for anaphylaxis with every dose.

- **The vaccine should be given in graded doses to patients with positive skin test under observation :**

- **0.05 ml 1:10 dilution. (same vaccine diluent, e.g ; normal saline)**
- **0.05 ml full strength.**
- **0.1 ml full strength.**
- **0.15 ml full strength.**
- **0.2ml full strength .**
- _____

Adapted from Kelso et al.[17],with permission from Elsevier Ltd.

Vaccine where the full dose was 0.5 ml ,physicians should give the following doses at 15 – 20 minute intervals. Patients under observation if an anaphylactic reaction should it occur.
Patients were observed for at least 30 minutes after the last dose.

Pitfalls In Vaccination Practice



If there was no scar or blister after BCG vaccination , should I revaccinate ?

A raised blister will appear in most people vaccinated with BCG , but not everyone.

If the child didn't have this reaction to the vaccine , it doesn't mean that they have not responded to it. There's no need to vaccinate with BCG a second time. (BCG vaccination scar , NHS).

But if we are uncertain whether a vaccine dose was given or there's no documentation , the dose in question should be given following a negative tuberculin skin test.

N.B :BCG can be given up to 3 months following a negative tuberculin skin test (The Green Book , immunization against infectious diseases)

What if my practice identifies a perinatally HBV-exposed newborn that did not receive HBIG before hospital discharge ?

The infant should receive an urgent referral to receive HBIG, which can be administered up to 7 days after birth. If more than 7 days have passed , HBIG is unlikely to be effective in preventing transmission.

- No efficacy was found if HBIG was given more than 7 days after birth and a significant decrease in efficacy was observed if it was given more than 48 hours after birth.

- **It's still important for the infant to complete the hepatitis B vaccine series, and providers should adhere to the minimum interval between doses.**

Bear in mind ; Post vaccination serologic testing (PVST) is recommended for infants and children born to women with hepatitis B infection. Serologic testing confirms whether the child has developed immunity or has been infected with HBV. PVST should include hepatitis B surface antigen (HBsAg) and hepatitis B surface antibody (anti-HBs) only. PVST should occur between 9-12 months of age or 1-2 months after vaccine series completion, if the series is delayed.

Note : Tests for antibodies to hepatitis B core antigen (anti-HBc) should not be ordered.

What is the best management of infants from mothers with perinatal chicken pox ?

Varicella zostor prophylaxis in newborn ;

Variations in guidelines…

Australian Society for infectious diseases

Treat infant according to timing of maternal chickenpox ;

Maternal chickenpox > 7 days before delivery :

- **No ZIG required**
- **No other interventions even if baby has chickenpox at or very soon after birth unless the infant is preterm < 28 weeks or low birth weight < 1000 gms**
- **No isolation of infant from mother**
- **Breast feeding encouraged**

Maternal chickenpox 7 days before to 2 days after delivery :

- **ZIG required immediately ; should be given < 24 hours after birth but may be given up to 72 hours**
- **No other interventions**
- **discharge term infants as soon as possible**
- **No isolation of infant from mother**
- **Breast feeding encouraged**

Maternal chickenpox > 2 – 28 days after delivery :

- **ZIG required ; if infant < 28 weeks gestation or < 1000 g birth weight give ZIG preferably within 96 hours but can be given up to 10 days post maternal rash**
- **Discharge term infants as soon as possible**
- **No isolation of infant from mother**
- **Breast feeding encouraged**
- **Some experts give ZIG to term babies > 2-8 days of age when mothers develop chickenpox but little data to support this**

N.B : intravenous immune globulin can be used if ZIG is not available.

Infant develops chickenpox ;

Very preterm infant in nursery :

IV aciclovir 20 mg/kg/dose every 8 hours as slow infusion (1-2 hours)

Term infant at home or on post natal ward :

Admit to paediatric unit

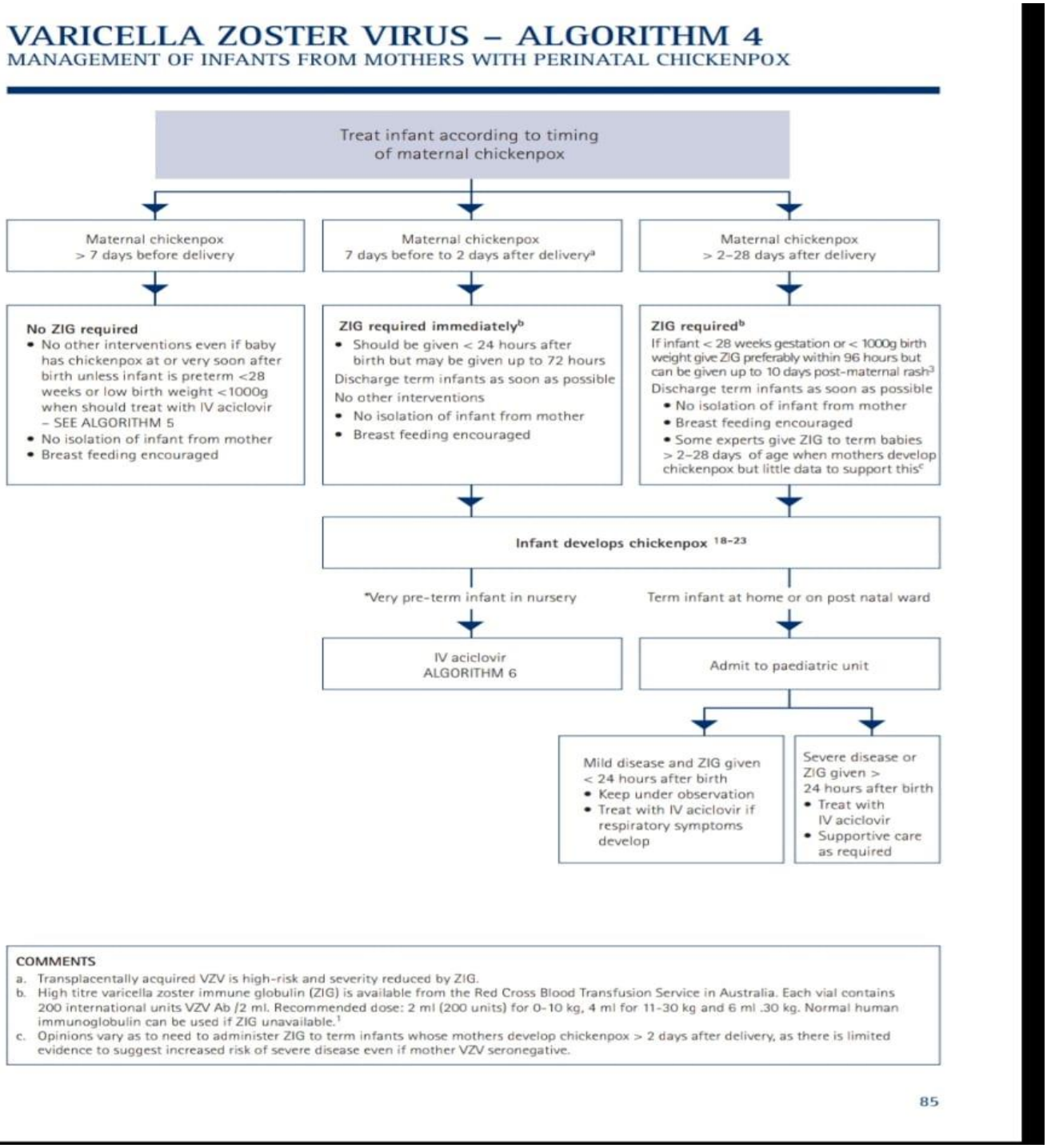
Mild disease and ZIG given < 24 hours after birth ;

- **Keep under observation**
- **Treat with IV aciclovir if repiratory symptoms develop**

Severe disease Or ZIG given > 24 hours after birth ;

- **Treat with IV aciclovir**
- **Supportive care as required**

Australian Society for infectious diseases



VARICELLA ZOSTER VIRUS – ALGORITHM 4

MANAGEMENT OF INFANTS FROM MOTHERS WITH PERINATAL CHICKENPOX

Treat infant according to timing of maternal chickenpox

Maternal chickenpox > 7 days before delivery

No ZIG required
- No other interventions even if baby has chickenpox at or very soon after birth unless infant is preterm <28 weeks or low birth weight <1000g when should treat with IV aciclovir – SEE ALGORITHM 5
- No isolation of infant from mother
- Breast feeding encouraged

Maternal chickenpox 7 days before to 2 days after delivery[a]

ZIG required immediately[b]
- Should be given < 24 hours after birth but may be given up to 72 hours

Discharge term infants as soon as possible

No other interventions
- No isolation of infant from mother
- Breast feeding encouraged

Maternal chickenpox > 2–28 days after delivery

ZIG required[b]

If infant < 28 weeks gestation or < 1000g birth weight give ZIG preferably within 96 hours but can be given up to 10 days post-maternal rash[3]

Discharge term infants as soon as possible
- No isolation of infant from mother
- Breast feeding encouraged
- Some experts give ZIG to term babies > 2–28 days of age when mothers develop chickenpox but little data to support this[c]

Infant develops chickenpox [18–23]

*Very pre-term infant in nursery → IV aciclovir ALGORITHM 6

Term infant at home or on post natal ward → Admit to paediatric unit

Mild disease and ZIG given < 24 hours after birth
- Keep under observation
- Treat with IV aciclovir if respiratory symptoms develop

Severe disease or ZIG given > 24 hours after birth
- Treat with IV aciclovir
- Supportive care as required

COMMENTS

a. Transplacentally acquired VZV is high-risk and severity reduced by ZIG.
b. High titre varicella zoster immune globulin (ZIG) is available from the Red Cross Blood Transfusion Service in Australia. Each vial contains 200 international units VZV Ab /2 ml. Recommended dose: 2 ml (200 units) for 0-10 kg, 4 ml for 11-30 kg and 6 ml .30 kg. Normal human immunoglobulin can be used if ZIG unavailable.[1]
c. Opinions vary as to need to administer ZIG to term infants whose mothers develop chickenpox > 2 days after delivery, as there is limited evidence to suggest increased risk of severe disease even if mother VZV seronegative.

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Royal college of obstetricians and gynecologists (RCOG)

- For babies born to mothers who have had chickenpox within the period of 7 days before to 7 days after delivery , it is therefore vital that the neonate receives prophylaxis as soon as possible with VZIG without antibody testing in infant.
  - If any symptoms develop despite VZIG prophylaxis , early treatment with IV Acyclovir 20 mg / kg / dose 8 hourly is recommended.

American Academy of Paediatrics , Red Book

Newborn infant whose mother had onset of chickenpox within 5 days before delivery or within 48 hours after delivery is a candidate for VZIG.

- VZIG given IM at a recommended dose of 125 iu/10kg (min dose 62.5 iu , up to a max of 625 iu)

- **If VZIG not available , IVIG 400 mg/kg administer once**
- **If VZIG not available or more than 96 hours have passed after the exposure , some experts recommend prophylaxis with IV Acyclovir 20 mg/kg/dose administer 4 times per day (max daily dose of 3200 mg) beginning 7 to 10 days after exposure.**
- **VZIG is not indicated for healthy full term infants who are exposed postnatally to varicella , including whose mother's rash developed more than 48 hours post delivery.**

Paediatric Protocol For Malaysian Hospital

Infants born to mothers who develop varicella between 7 days antenatally and 14 days postnatally should receive as prophylaxis :

1. **Varicella Zoster immunoglobulin (VZIG) within 96 hours of initial exposure (attenuation of disease might still be achieved with administration of VZIG up to 10 days).**
2. **If VZIG not available , IVIG 400 mg/kg AND**
3. **IV Acyclovir 15 mg/kg/dose over 1 hour every 8 hours (total 45 mg/kg /day) for 5 days.**

Varicella Zoster Prophylaxis in Newborn : Variations in Guidelines

Paediatric Protocol For Malaysian Hospital 3rd Edition, 2012.	Infants born to mothers who develop varicella between 7 days antenatally and 14 days postnatally should receive as prophylaxis: 1. **Varicella Zoster immunoglobulin (VZIG)** within 96 hours of initial exposure (Attenuation of disease might still be achieved with administration of VZIG up to 10 days) . **NOT AVAILABLE IN HUKM** 2. If VZIG not available, **IVIG 400mg/kg AND** 3. **IV Acyclovir 15mg/kg/dose over 1 hour every 8 hours** (total 45mg/kg/day) for 5 days. Infants whose mothers develop Zoster before or after delivery have maternal antibodies and they will not need VZIG.
The Royal College of Obstetricians and Gynaecologists (RCOG), 2015	For babies born to mothers who have had chickenpox within the period 7 days before to 7 days after delivery, it is therefore vital that the neonate receives **prophylaxis as soon as possible with VZIG** without antibody testing in infant. **If any symptoms develop** despite VZIG prophylaxis, **early treatment with IV Acyclovir 20mg/kg/dose 8 hourly** is recommended [6,7]
Red Book 29th Edition, American Academy of Paediatric, 2012	Newborn infant whose mother had onset of chickenpox within 5 days before delivery or within 48 h after delivery is a candidate for VZIG. **VZIG (VariZIG®):** given IM at recommended dose of **125iu/10kg,** (min dose 62.5iu, up to a max of 625iu) If VZIG not available, **IVIG 400mg/kg administer once** If VZIG not available or more than 96 hours have passed after the exposure, some experts recommend prophylaxis with **IV Acyclovir 20mg/kg/dose** administer 4 times per day (max daily dose of 3200mg) beginning 7 to 10 days after exposure. (Varicella can develop between 2 and 16 days after birth in infants born to mothers with active varicella around the time of delivery) VZIG is not indicated for healthy full term infants who are exposed postnatally to varicella, including infants whose mother's rash developed more than 48h post delivery.
BNF for Children, Sept 2016-2017	**VZIG (non-proprietary):** 250mg to be administered IM as soon as possible-not later than 10 days after exposure, 2nd dose to be given if further exposure occurs more than 3 weeks after the first dose, no evidence that it is effective in severe disease. Prophylactic **IV Acyclovir 10mg/kg every 8 hours** should be considered for neonates whose mothers develop chicken pox 4 days before to 2 days after delivery and to be continued until serological tests confirm absence of virus.

References:

1. Kett, J. (2013). Perinatal Varicella. Pediatrics in Review, American Academy of Pediatrics. 34(1), pp.49-51.
2. Imam, H. Thomas, T. Hoong, N T. Perinatally Acquired Varicella. Paediatric Protocols for Malaysian Hospitals 3rd Edition. Ministry of Health, Malaysia.
3. Speer, M. (2017). Varicella-zoster Infection in the Newborn. UpTpDate. [Accessed on 14th June 2018].
4. Lamounier, J., Moulin, Z. and Xavier, C. (2004). Recommendations for breastfeeding during maternal infections. Jornal de Pediatria, 80(8), pp.181-188.
5. Green-top Guideline No. 13(2015). Chicken Pox in Pregnancy. Royal College of Obstetricians and Gynaecologists. London, United Kingdom.
6. Sharland M, Cant A, Davies EG, Elliman DA, Esposito S, Finn A, et al., editors. Chicken Pox—Varicella Zoster. In: Manual of Childhood Infections. The Blue Book. Oxford: Oxford University Press; 2011. p. 84, 813.
7. Public Health England. Varicella: the green book, chapter 34. London: Public Health England; 2012 [https://www.gov.uk/government/publications/ varicella-the-green-book-chapter-34]. P 476
8. Red Book (2012). 2012 Report of the Committee on Infectious Disease 29th Edition. American Academy of Pediatrics.

Varicella may be a severe illness in patients with nephrotic syndrome receiving corticosteroids or other immunosuppressive drugs !

- Susceptible patients (those unimmunized or with no history of varicella) who are exposed to a case of chickenpox should , therefore , receive a single dose of varicella zoster immunoglobulin , within 96 h of exposure to prevent or lessen the severity of the disease. Since , this preparation is expensive and not easily available , a single dose of intravenous immunoglobulin (400 mg /kg) may be used instead.
- Patients who develop varicella should receive intravenous acyclovir (1500 mg/m2 / day in three doses) or oral acyclovir (80 mg/kg/day in four doses) for 7-10 days. The dose of prednisolone should be tapered to 0.5 mg/kg/day or lower during the infection.

(Indian journal of nephrology)

Can someone get tetanus from something in their house ?

Spores of tetanus bacteria are everywhere in the environment.

If there is a wound , even minor. Tetanus spores may be present on any surface where dust can be.

Even a scratch from a thorn or an animal , a splinter , bug bite , a stab from an unsterile sewing needle can lead to invasion of tetanus bacteria. The source of infection can be indoors or out – in a home , yard , garden , farm or elsewhere. (The New York Times)

If your skin gets pierced from anything , be it your own kitchen knife or a rusty old screw , it's worth making sure that you are up to date with tetanus vaccination.







Wound management and tetanus prophylaxis (UpToDate)

Previous Doses of tetanus toxoid	Clean and minor wound		All other wounds	
	Tetanus toxoid – Containing vaccine	**Human tetanus immune globulin**	**Tetanus toxoid – Containing vaccine**	**Human tetanus immune globulin**
< 3 doses Or Unknown	Yes	No	Yes	Yes
≥ 3 doses	Only if last dose given **≥ 10 years ago**	No	Only if last dose given **≥ 5 years Ago**	No

Prophylaxis should be administered as soon as possible following wound or traumatic injury
and even upon delayed presentation due to long and variable incubation period of tetanus (3 days – 21 days)

- Human tetanus immune globulin : 250 units intramuscularly at a different site than tetanus toxoid ; intravenous immune globulin should be administered (200 to 400 mg /kg)if human tetanus immune globulin is not available. The incubation period can be as short as 3 days or as long as 21 days. For patients who have been vaccinated against tetanus previously but who are not up to date , there is likely to be little benefit in administering human tetanus immune globulin more than 1 week or so after the injury. However , for patients thought to be completely unvaccinated , human tetanus immune globulin should be given up to 21 days following the injury.
- Equine tetanus antitoxin should no longer be used , as there is a risk of hypersensitivity and serum sickness up to 10 days after injection (MSF medical guidelines 2022).
- In immunocompromised patients , tetanus immune globulin for tetanus-prone wounds should be used in conjunction with tetanus-diphtheria booster or primary series (in patients who have not completed a primary series or whose vaccination has expired.)

Wound management and tetanus prophylaxis in newborns :

The minimum age of the first dose of tetanus toxoid containing vaccine is 6 weeks .

- **Immunization of pregnant women** ensures a 100 % passive protection of newborns against tetanus before administration of the first vaccine doses.

- **The recommended timing** for maternal Tdap vaccination is between **27 weeks and 36 weeks** of gestation , to maximize the maternal antibody response and passive antibody transfer and levels in the newborn. However , **the Tdap vaccine maybe safely given at any time** during pregnancy if needed for wound management. **(ACOG Guidelines)**
  - In pregnant women , administer **at least 2 doses** before delivery ; the first dose as soon as possible and **the second dose at least 4 weeks after the first** and at least **2 weeks before delivery (MSF Medical Guidelines 2022)**
  - **The Centers for Disease Control and Prevention recommends all women receive a Tdap vaccine during the 27th through 36th week of each pregnancy.**

Does human bite need a tetanus shot ?



YES , **Tetanus is a serious , potentially life-threatening infection that can be transmitted by an animal or human bite. People who are bitten should receive a tetanus vaccine if they are not up to date with tetanus vaccination.**

Does cat scratch need anti rabies post exposure prophylaxis ?

Rabies is a viral disease that affects the central nervous system of mammals. All mammals are susceptible to the rabies virus

According to WHO Classification of Rabies Post-Exposure Prophylaxis Modalities:

Minor scratches or abrasions without bleeding are classified as category II exposure that requires anti rabies vaccine (0, 3 , 7 , 14 , 28)

Vaccine should be injected as soon as possible

Single or multiple transdermal bites or scratches are classified as category III exposure that requires anti rabies vaccine and immunoglobulin

Vaccine and immunoglobulin should be administered at distant sites as soon as possible.

Bear in mind ;

- Immunogobulin can be administered up to 7 days after injection of the first dose of vaccine. If delayed beyond this time, rabies IG should not be given , as it may interfere with the active production of antibodies that usually become detectable.

- If a person has been previously fully vaccinated against rabies (in the last 5 years) only 2 doses of vaccine are given on day 0 and day 3 and rabies immunoglobulin should not be administered due to anamnestic response (rapid production of antibodies because of previous vaccination)

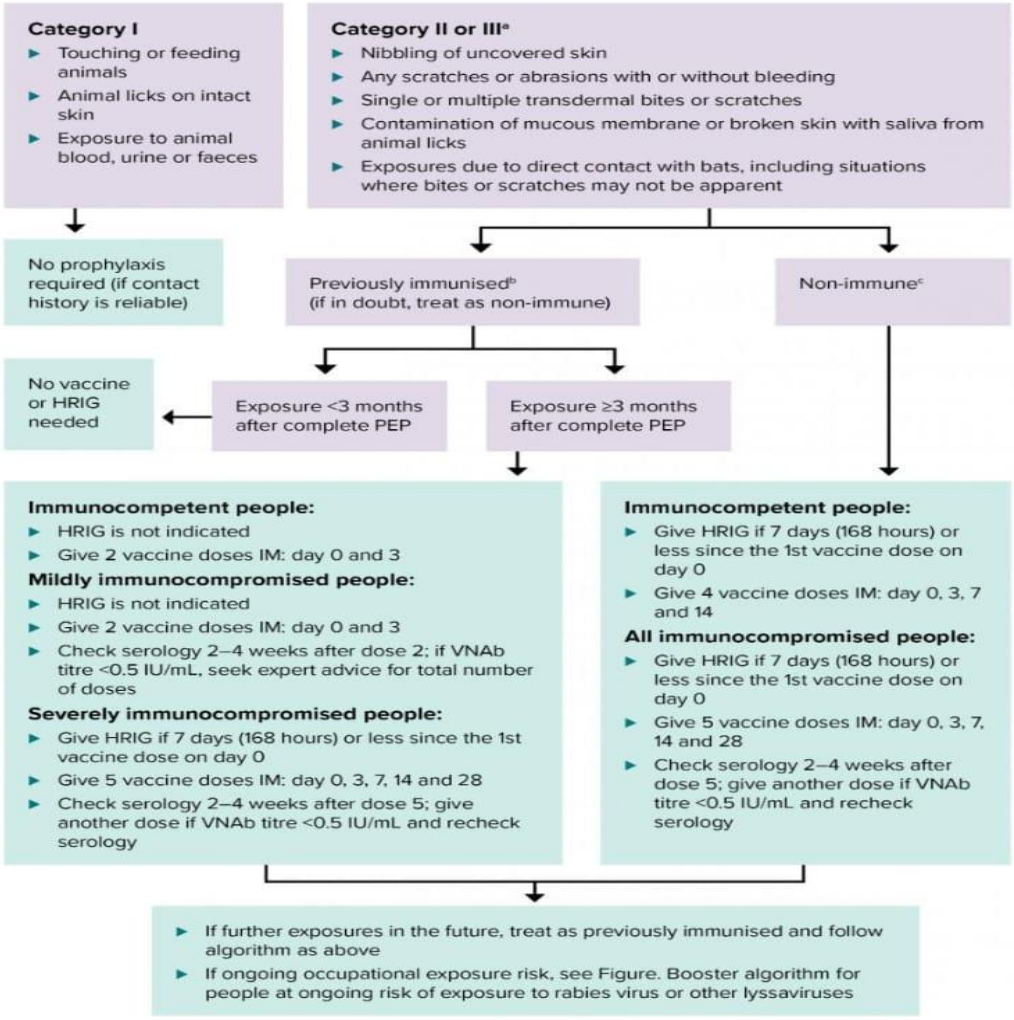
N.B : People who are scratched should receive a tetanus vaccine if they are not up to date with tetanus vaccination. it's worth making sure that you are up to date with tetanus vaccination.

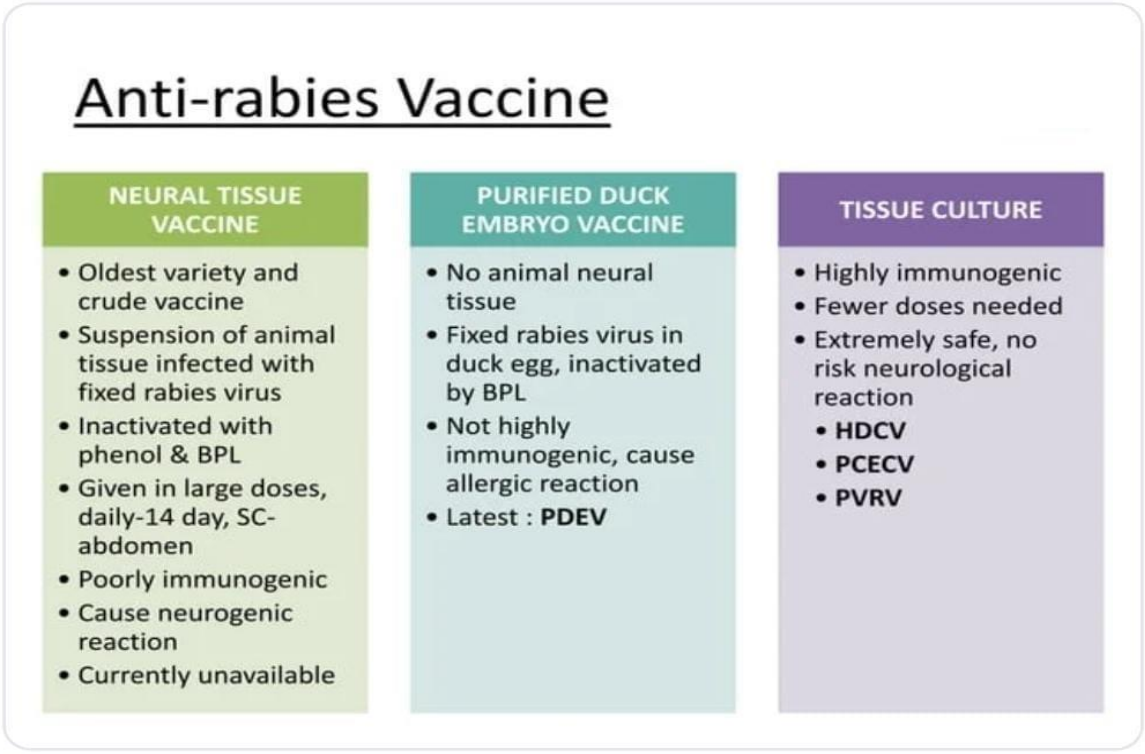
Rabies Post-Exposure Prophylaxis Modalities

In countries or areas enzootic for rabies , exposure to suspected or confirmed rabid animals are categorized by WHO as follows :

Category of exposure	Description	Post-exposure prophylaxis
Category I	Touching or feeding animals, licks on intact skin with secretions or excretion of rabid animal or person	Not regarded as exposures, therefore no PEP required.
Category II	Nibbling of uncovered skin, minor scratches or abrasions without bleeding	Vaccine should be injected as soon as possible.
Category III	Single or multiple transdermal bites or scratches, licks on broken skin, contamination of mucous membrane with saliva from licks & exposure to bats	Vaccine and rabies immunoglobulin should be administered up to 7 days after injection of the first dose of vaccine

RABIES PEP (Australian immunization handbook)





Purified Duck Embryo Vaccine (PDEV) , 1 ml

Cell Culture Vaccines :

- Human Diploid Cell Vaccine (HDCV) , 1 ml
- Purified Chick Embryo Cell Vaccine (PCECV), 1 ml
- Purified Vero Cell Rabies Vaccine (PVRV) , 0.5 ml

The major advantage of third-generation rabies vaccines prepared in non-neural tissue cultures is the greatly reduced risk of neurological complications.

Indications For Use Of Rabies Immunoglobulin :

1. **All category III bites (WHO classification)**
2. **Licks on mucous membranes by wild or pet animals**
3. **Category II & III bites in case of AIDS patients with grossly reduced CD4 count**
4. **Varicella (can cause transient immunodeficiency in children)**
5. **Patients on long term Corticosteroids**
6. **Long term Chemotherapy**
7. **Radiation Therapy**
8. **Wound suturing (pre-infiltration)**

Note: Sutures (stitches)- Some wounds can be sutured within several hours of the injury, after the wound is thoroughly cleaned. Wounds to the face are usually closed immediately to avoid developing a scar. However , due to risk of infection , some bite wounds may not be sutured immediately. These wounds may be left open & sutured 72 hours after the injury. When wounds are left open , they are flushed with saline , left open to drain , covered with a dressing ,& examined daily to look for signs of infection.

Immediate suturing may NOT be recommended for wounds at high risk of becoming infected , including :

- **Crush injuries**
- **Puncture wounds**
- **Bites involving the hands**
- **Dog bite wounds that occurred many hours earlier**
- **Cat or human bites, except those to the face**
- **Bite wounds in people who have a weakened immune system.**

Bear in mind ;

- Animal bites especially on the head , neck face , hands and genitals are category III exposures because of rich innervation of these areas. (World Health Organization)
- Regarding Vaccination status of the biting animal ; Although unvaccinated animals are more likely to transmit rabies , vaccinated animals can also do so if the vaccination of the biting animal was ineffective for any reason. A History of rabies vaccination in an animal is not always a guarantee that the biting animal is not rabid. Animal vaccine failures may occur because of improper administration or poor quality of the vaccine , poor health status of the animal and the fact that one vaccine dose doesn't always provide long-lasting protection against rabies infection in dogs/cats . Hence , appropriate documentation of vaccination status of dog/cat and proper history should be elicited before deciding to defer post-exposure prophylaxis after bite by vaccinated dog/cat. (Indian protocol for rabies PEP)
- If you have been exposed to rabies virus, you should get vaccinated regardless of concurrent illnesses , fever , pregnancy , breastfeeding , or weakened immune system.(CDC)
- Exposures incurred in the course of caring for humans with rabies could theoretically transmit the infection. No cases of rabies acquired in this way has been documented , but Post Exposure Prophylaxis (PEP) should be considered for exposed individuals. (Protocol for rabies PEP, Ontario , Canada)
- Due to the potentially long incubation period for rabies , there is no time limit for giving rabies PEP. Rabies should ideally be provided as soon after exposure as possible. When patients present well after the exposure event, regard the first day of presentation as day 0 for vaccine administration.
- Day 0 – Day of first dose of vaccine given , not the day of bite.
- All modern Tissue Cultures Vaccines (TCVs) are equally effective and safe.
- Never inject the vaccines into the gluteal region.
- Interchange of vaccines acceptable in special circumstances but not to be done routinely.
- Reconstituted vaccine to be used immediately.
- Vaccine dosage is same for all age groups

Deviations from Recommended Rabies Post-Exposure Prophylaxis Regimen :

- Once the decision to initiate rabies PEP has been made , the PEP regimen should be started as soon as possible.
- Every effort should be made to adhere to the recommended PEP regimen schedule , especially the first two days of treatment , Days 0 and 3. After Day3 of the regimen , deviations of a few days are acceptable.
- For most minor delays or interruptions, the vaccination schedule can be shifted and resumed as though the patient was on schedule.
- For example , if a patient misses the dose scheduled for Day 7 and presents for vaccination on Day 10,the day 7 dose should be administered that day, and the next dose given one week later on Day 17
- When substantial deviations from the schedule occur , immune status should be assessed by performing serologic testing 7-14 days after administration of the final dose in the series (Documented rabies virus neutralizing antibody titres {VNA} of at least 0.5 IU /ml.)

Human and animal bites : antimicrobial prescribing : NICE Guidelines

Prophylactic antibiotics

- **Human bite**

Offer antibiotic prophylaxis if :

- **Broken the skin & drawn blood**
- **Broken the skin of hands , feet , face , genitals , overlying cartilaginous structures or area of poor circulation**
- **patient at risk of serious wound infection (e.g immunocompromised)**
- **bite ≤ 72 hr old , even if no sign of infection**

- **Cat bite**

Offer antibiotic prophylaxis if :

- **Broken the skin & drawn blood**
- **deep bite**

Prophylactic antibiotics

- **Dog and other animal bites**

Offer antibiotic prophylaxis if :

- **If wound ≤ 48 hr old , if broken the skin and drawn blood AND if :**
  - **bites to hand , foot , face or genitalia**
  - **Puncture wounds**
  - **wounds requiring surgical debridement**
  - **crush wounds with devitalized tissue**
  - **wounds with associated oedema**
  - **wounds involving joints , tendons , ligaments , or suspected fractures**
  - **wounds that have undergone primary closure**
  - **patients at risk of serious wound infection (immunosuppressed)**
  - **asplenic patients , even after trivial animal bites**
  - **patients with prosthetic implants e.g heart valve , VP shunt**
- **Co-amoxiclav 3 days prophylaxis or 5 days if appears infected**
  - **If unable to take orally give co-amoxiclav IV**
- **Penicillin allergy**
  - **Aged > 12 yr : doxycycline with metronidazole**
  - **aged < 12 yr : co-trimoxazole**
  - **if unable to take oral antibiotics : cefuroxime with metronidazole**
- **Antibiotics not generally required if wounds ≥ 2 days and no sign of local or systemic infection.**

Choice of antibiotic for prophylaxis and treatment : children & young people under 18 years

Prophylaxis and treatment	Antibiotic, dosage & course length for prophylaxis (3 days) and treatment (5days)
Choice for children under 1 month	Seek specialist advice
First-choice oral antibiotic for children aged 1 month and over	Co-amoxiclav : 1 month to 11 months:0.25ml/kg of 125/31 suspension three times a day 1 year to 5 years:0.25ml/kg or 5 ml of 125/31 suspension three times a day 6 years to 11 years : 0.15 ml/kg or 5 ml of 250/62 suspension three times a day 12 years to 17 years : 250/125 mg or 500/125 mg three times a day Co-amoxiclav 400/57 suspension may also be considered to allow for twice-daily dosing
Alternative first-choice oral antibiotic for children under 12 years for penicillin allergy or if co-amoxiclav is unsuitable	Co-trimoxazole (off-label use; see the BNF for Children for information on monitoring) : 6 weeks to 5 months : 120 mg or 24 mg/kg twice a day 6 months to 5 years , 240 mg or 24 mg/kg twice a day 6 years to 11 years , 480 mg or 24 mg/kg twice a day For off-label use , follow relevant professional guidance , taking full responsibility for the decision. Informed consent should be obtained and documented. See the General Medical Council's good practice in prescribing and managing medicines and devices for information.
Alternative first-choice oral antibiotics for young people aged 12 to 17 years for penicillin allergy or if co-amoxiclav is unsuitable	Doxycycline : 200 mg on first day, then 100 mg or 200 mg daily With metronidazole : 400 mg three times a day
Alternative first-choice oral antibiotics in pregnancy for penicillin allergy or if co-amoxiclav is unsuitable	Seek specialist advice
First-choice intravenous antibiotic (if unable to take oral antibiotics or severely ill)	Co-amoxiclav : 1 month to 2 months : 30 mg/kg twice a day 3 months to 17 years : 30 mg/kg three times a day (maximum per dose 1.2 g)
Alternative first-choice intravenous antibiotics for penicillin allergy or if co-amoxiclav is unsuitable If a cephalosporin is not appropriate, seek specialist advice	Cefuroxime (caution in penicillin allergy): 1 month to 17 years : 20 mg/kg three times a day (maximum 750 mg per dose), which can be increased to 50 mg/kg to 60 mg/kg three or four times a day (maximum per dose 1.5 g) With metronidazole : 1 month : loading dose 15 mg/kg , then (after 8 hours) 7.5 mg/kg three times a day 2 months to 17 years : 7.5 mg/kg three times a day (maximum per dose 500 mg) Ceftriaxone (caution in penicillin allergy): 1 month to 11 years (up to 50 kg) : 50 mg/kg to 80 mg/kg once a day (maximum 4 g per day) 9 years to 11 years (50 kg and above) and 12 years to 17 years : 1 g to 2 g once a day With metronidazole : 1 month : loading dose 15 mg/kg , then (after 8 hours) 7.5 mg/kg three times a day 2 months to 17 years : 7.5 mg/kg three times a day (maximum per dose 500 mg)

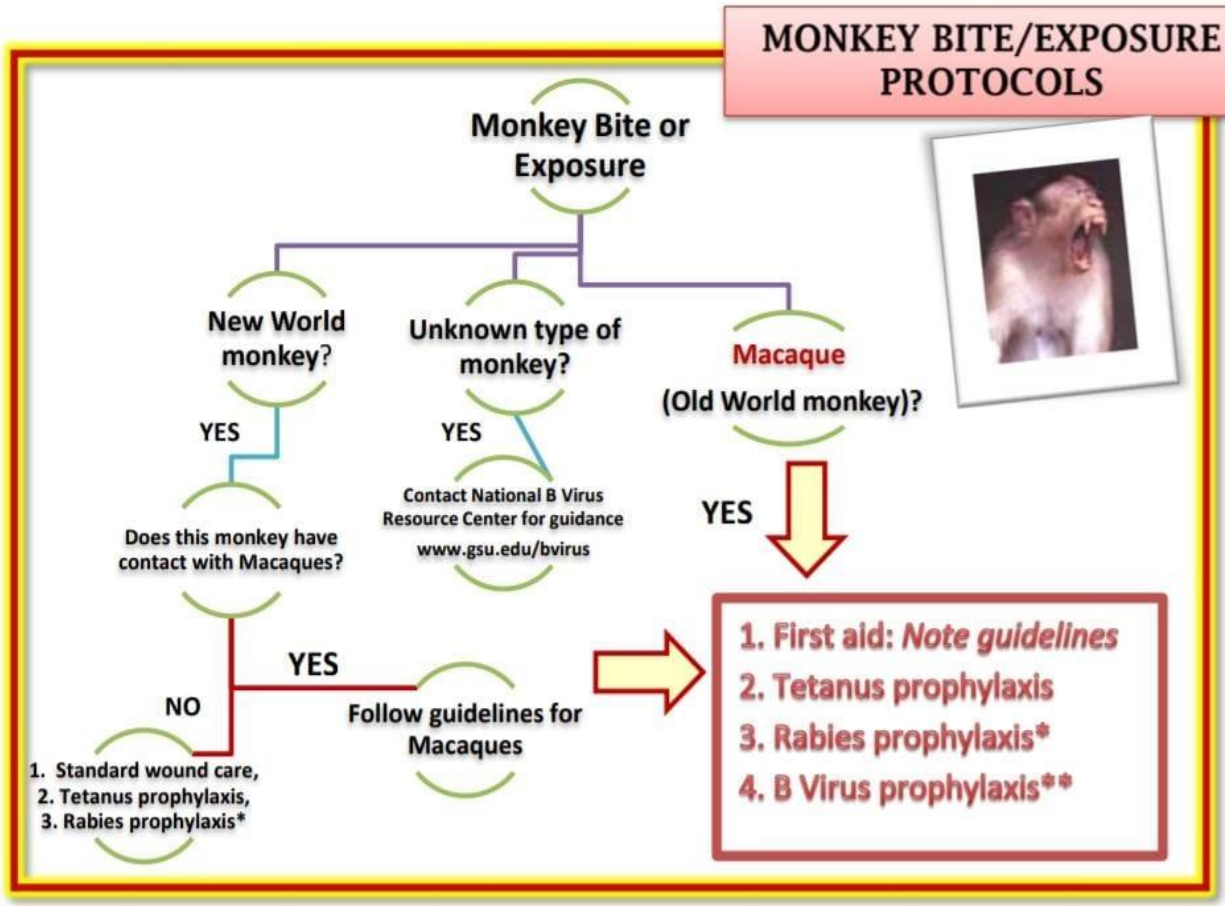
See the BNF and summary of product characteristics for appropriate use and dosing in specific populations , for example , for hepatic or renal impairment , in pregnancy , when breastfeeding and when administering intravenous (or, if appropriate , intramuscular) antibiotics. A 5-day course is appropriate for treating most human or animal bites , but course length can be increased to 7 days (with review) based on clinical assessment of the wound , for example , if there is significant tissue destruction or it has penetrated bone, joint , tendon or vascular structures.

Choice of antibiotic for prophylaxis and treatment : adult aged 18 years & over

Prophylaxis and treatment	Antibiotic, dosage & course length for prophylaxis (3 days) and treatment (5days)
First choice oral antibiotic	Co-amoxiclav : 250/125 mg or 500/25 mg three times a day
Alternative first-choice oral antibiotics for penicillin allergy or if co-amoxiclav is unsuitable	Doxycycline : 200 mg on first day , then 100 mg or 200 mg daily With Metronidazole : 400 mg three times a day
Alternative first-choice oral antibiotics in pregnancy for penicillin allergy or if co-amoxiclav is unsuitable	Seek specialist advice
First-choice intravenous antibiotic (if unable to take oral antibiotics or severely unwell)	Co-amoxiclav : 1.2 g three times a day
Alternative first-choice intravenous antibiotics for penicillin allergy or if co-amoxiclav is unsuitable If a cephalosporin is not appropriate, seek specialist advice	Cefuroxime (caution in penicillin allergy): 750 mg three times a day (increased to 750 mg four times a day or 1.5 g three or four times a day if infection is severe) With Metronidazole : 500 mg three times a day Ceftriaxone (caution in penicillin allergy) 2 g once a day With Metronidazole : 500 mg three times a day

See the BNF and summary of product characteristics for appropriate use and dosing in specific populations , for example , for hepatic or renal impairment , in pregnancy , when breastfeeding and when administering intravenous (or, if appropriate , intramuscular) antibiotics. A 5-day course is appropriate for treating most human or animal bites , but course length can be increased to 7 days (with review) based on clinical assessment of the wound , for example , if there is significant tissue destruction or it has penetrated bone, joint , tendon or vascular structures.

MONKEY BITE/EXPOSURE PROTOCOLS



FIRST AID GUIDELINES FOR MACAQUE BITE/EXPOSURE

Cleanse the exposed area by thoroughly washing and scrubbing the area or wound with soap, concentrated solution of detergent, povidone-iodine, or chlorhexidine and water, and then irrigate the washed area with running water for 15-20 minutes.

WARNING: a specimen for PCR testing should not be obtained from the wound area prior to washing the site because it could force virus more deeply into the wound, reducing the effectiveness of the cleansing protocol. After the site is cleansed, a serum specimen should be obtained from the patient to provide a baseline antibody level. (See Specimen Collection and B virus Detection)

**RABIES PROPHYLAXIS INFORMATION ON REVERSE NOTE: SPECIFIC ANTIVIRAL DRUGS/DOSES ON REVERSE*

**B virus antiviral Therapy

- **Administration of B virus – specific immunoglobulin , if available , may be effective , and ganciclovir or acyclovir may be effective. The proposed use of specific immunoglobulin is based solely on the effectiveness of i.v. immunoglobulin as a safe prophylactic against other alphaherpesvirus infections, notably varicella-zoster virus.**
- **Recommended dosages for specific antivirals are as follows :**

Prophylaxis for exposure to B virus (adults)

- **Valacyclovir - 1 g by mouth every 8 hours for 14 days , or**
- **Acyclovir – 800 mg by mouth 5 times daily for 14 days**

Treatment of B virus infection

Antiviral therapy has been recognized as an effective prevention of infection progression in humans and animal trials as well , when administered sufficiently early after exposure. Acyclovir and the related family of nucleoside analogues were noted to be effective when given in high doses , e.g, acyclovir at 10-15 mg /kg three times daily for 14-21 days.

With no CNS symptoms

- **Acyclovir- 12.5- 15 mg/kg intravenously every 8 hours, or**
- **Ganciclovir- 5 mg/kg intravenously every 12 hours**

With CNS symptoms

- **Ganciclovir- 5 mg/kg intravenously every 12 hours**

Splenectomy vaccination guideline,
When to give vaccines ?

- **Elective splenectomy : immunise at least TWO weeks(ideally four to six weeks) before surgery.**
- **Emergency splenectomy : immunize at least TWO weeks post surgery or when the patient is feeling sufficiently well.**

In- hospital vaccination protocol :

Administer vaccination on the day of discharge or day 14 , whichever comes first

- **(PCV 13 – Prevnar 13) 0.5 ml IM**
- **Haemophilus influenza type b vaccine (HIBERIX) 0.5 ml IM**
- **Meningococcal conjugate quadrivalent vaccine (Menactra) 0.5ml IM**
- **Meningococcal serogroup B (Bexsero) 0.5 ml IM**

Follow-up vaccination needed ;

2 month follow up after initial vaccination

- **Pneumococcal polysaccharide (PPSV 23 – Pneumovax 23) 0.5 ml IM**
- **Meningococcal conjugate quadrivalent vaccine (Menactra or Nimenrix) 0.5ml IM**
- **Meningococcal serogroup B (Bexsero) 0.5 ml IM (> 1 month after first dose)**

Bear in mind :

- **Age 2-5 years with incomplete series (less than 3 doses of PCV13) : will need a second dose of PCV13 , 8 weeks after the most recent dose.**
- **Age 12- 59 months (unvaccinated or only 1 dose before age 12 months) : will need a second dose of Hib , 8 weeks after the most recent dose.**

Long-term follow up

- **Pneumococcal polysaccharide (PPSV 23 – Pneumovax 23) 0.5 ml IM 5 years after the first dose of this vaccine**
- **Meningococcal conjugate quadrivalent vaccine (Menactra or Nimenrix) 0.5ml IM recommended every 5 years**
- **Seasonal influenza vaccine is indicated annually**

In a person with functional or anatomic asplenia ;

Menactra shouldn't be administered on the same day as Prevnar 13

Studies have shown that co-administration may result in a decreased immune response to some of pneumococcal serotypes. If both vaccines are required , prevnar 13 should be administered first followed by Menactra at least 4 weeks later .

If these vaccines are inadvertently administered on the same day , Prevnar 13 should be repeated a minimum of 8 weeks later.

Menveo and Nimenrix may be co-administered with prevnar 13 .

(The Australian immunization Handbook)

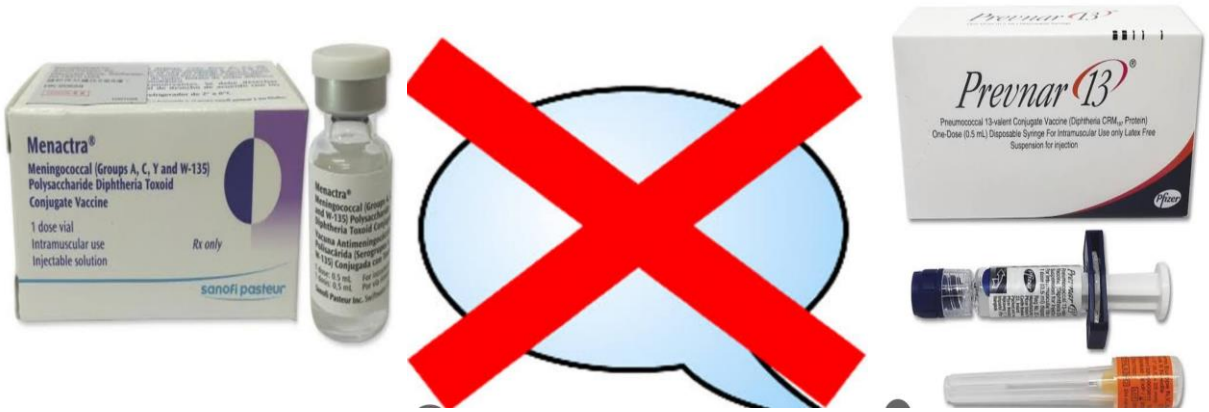


Table 2 Recommended immunization schedule for sickle cell disease

	Pneumococcus	Meningococcus and *Haemophilus influenzae* type b	Influenza
Under 2 years	Routine[a]	Routine[a]	Annual
Age 2–5 years (fully immunized)	Single dose PPV	Booster dose given as the Hib/MenC vaccine	Annual
Age 2–5 years (unvaccinated or partially vaccinated)	Two doses of PCV given 2 months apart, followed 2 months later by PPV	Two doses of the Hib/MenC vaccine given 2 months apart	Annual
Age >5 years (fully vaccinated)	Single dose PPV	Two doses of the Hib/MenC given 2 months apart	Annual
Age >5 years (unvaccinated)	Single dose PPV	Two doses of the Hib/MenC vaccine given 2 months apart	Annual
Reinforcing immunization	PPV every 5 years	MenC vaccine every 5 years, Hib vaccine not currently recommended	Annual

Note: Schedule summarized from Salisbury D, Ramsay M, Noakes K. Immunization against infectious diseases (The Green Book). London: Department of Health; 2006.[78] National childhood immunization in most developing countries does not currently include Hepatitis B, Hib or meningococcal vaccines. These should be strongly encouraged for children with SCD. In endemic settings consideration should also be given for immunization with typhoid vaccine.

Hib, *Haemophilus influenzae* type b; MenC, meningococcus group C; PCV, pneumococcal conjugate vaccine; PPV, pneumococcal polysaccharide vaccine.

a Routine UK immunization schedule.

Guidelines on Post-Exposure Prophylaxis for Measles :Assessment and treatment of infants (UK Health Security Agency)

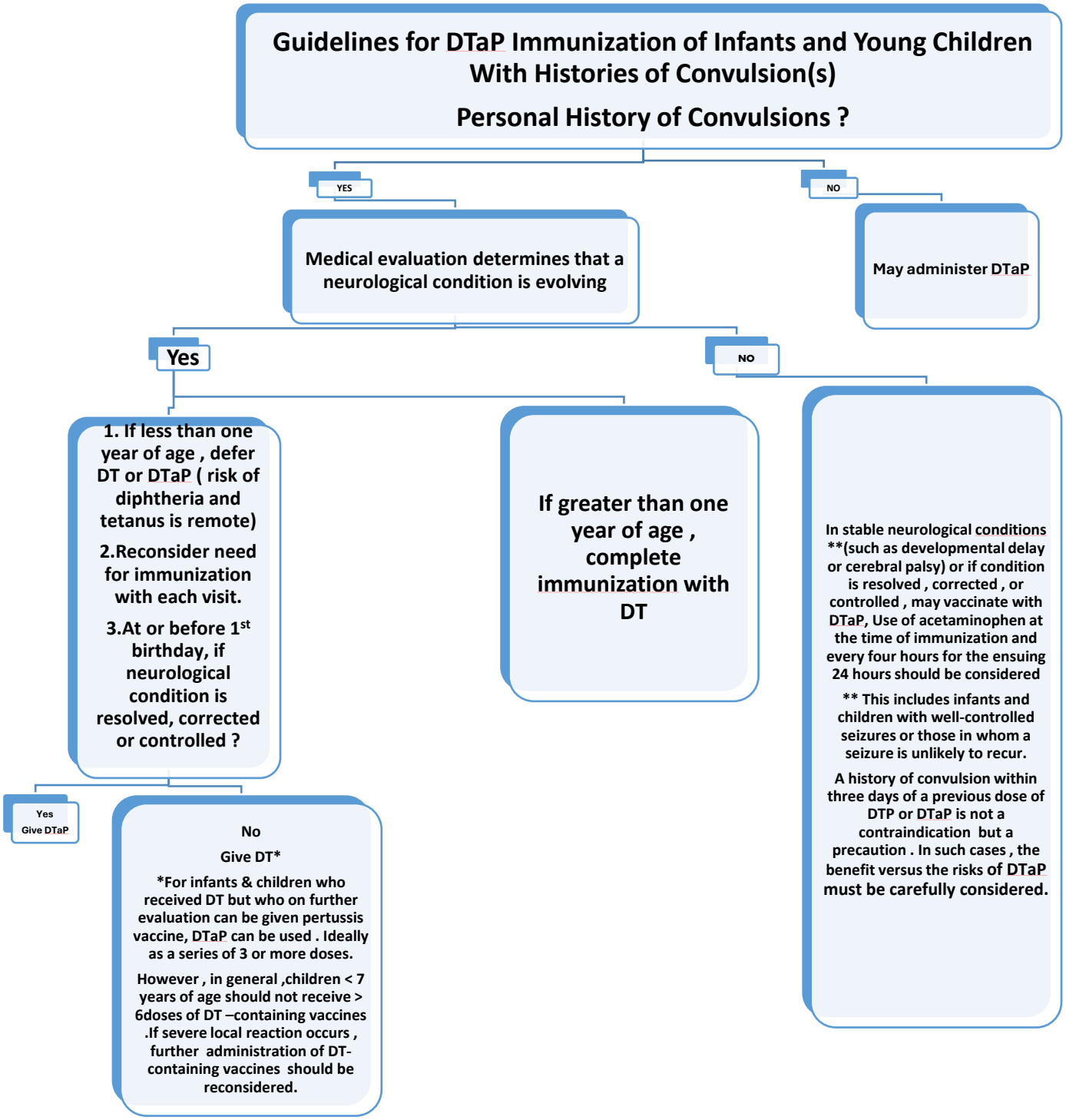
Infants		
Infants < 6 months	Assume susceptible and administer HNIG (Human Normal Immunoglobulin) , ideally within 72 hours but up to 6 days , regardless of maternal status	
Infants aged 6-8 months	For household exposure, Administer HNIG , ideally within 72 hours but up to 6 days if necessary	For exposure outside of the household , Administer MMR , ideally within 72 hours
Infants ≥ 9 months	Administer MMR vaccine , ideally within 72 hours of exposure	

HNIG (Human normal immunoglobulin) ; The dose of IMIG for these contacts is 0.5 mL/kg to a maximum of 15 mL and the dose of IVIG is 400 mg/kg.

Guidelines for DTaP Immunization of Infants and Young Children With Histories of Convulsion(s)

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(Los Angeles Department of Public Health of Acute Communicable Disease Control /Immunization Program).


Guidelines for DTaP Immunization of Infants and Young Children With Histories of Convulsion(s)
Personal History of Convulsions ?
YES
NO
Medical evaluation determines that a neurological condition is evolving
May administer DTaP
Yes
NO
1. If less than one year of age , defer DT or DTaP (risk of diphtheria and tetanus is remote)
2.Reconsider need for immunization with each visit.
3.At or before 1st birthday, if neurological condition is resolved, corrected or controlled ?
If greater than one year of age , complete immunization with DT
In stable neurological conditions **(such as developmental delay or cerebral palsy) or if condition is resolved , corrected , or controlled , may vaccinate with DTaP, Use of acetaminophen at the time of immunization and every four hours for the ensuing 24 hours should be considered
** This includes infants and children with well-controlled seizures or those in whom a seizure is unlikely to recur.
A history of convulsion within three days of a previous dose of DTP or DTaP is not a contraindication but a precaution . In such cases , the benefit versus the risks of DTaP must be carefully considered.
Yes
Give DTaP
No
Give DT*
*For infants & children who received DT but who on further evaluation can be given pertussis vaccine, DTaP can be used . Ideally as a series of 3 or more doses.
However , in general ,children < 7 years of age should not receive > 6doses of DT –containing vaccines .If severe local reaction occurs , further administration of DT-containing vaccines should be reconsidered.

Epilepsy and vaccinations: Italian guidelines

Recommendation 1 :

Vaccinations should be performed without contraindication in children with previous febrile seizures (class of evidence III ; strength of recommendation A)

Recommendation 2 :

The risk of febrile seizures shouldn't discourage parents from vaccinating their children (class of evidence III ; strength of recommendation A)

Note : Parents should be informed that some vaccines could be associated with febrile reactions with consequent seizures , in particular in children with personal history of febrile seizures and/or younger than 6 years.

Recommendation 3 :

Vaccinations should be performed without contraindication in children with idiopathic or symptomatic epilepsy (class of evidence III ; strength of recommendation A)

Recommendation 4 :

The risk of epilepsy shouldn't discourage parents from vaccinating their children (class of evidence III ; strength of recommendation A)

Recommendation5:

Vaccinations should be performed without contraindication in children with epileptic encephalopathies (Dravet syndrome , West syndrome) (class of evidence III ; strength of recommendation A)

Recommendation 6:

The possible risk of epileptic encephalopathy shouldn't discourage parents from vaccinating their children (class of evidence III ; strength of recommendation A)

From The critical appraisal of the literature, the following indications can be drawn :

(1) Some vaccinations , in particular DTP (especially whole-cell vaccine) and MMR (especially combined with varicella) , can cause fever with possible febrile seizures.

(2) The rate of febrile seizures is similar in children with and without a personal history of previous febrile seizures.

(3) Vaccine-induced febrile seizures are no more frequent than febrile seizures with any other cause of fever.

(4) The risk of nonfebrile seizures following vaccine-induced febrile seizures is not higher than in children who have not shown vaccine-induced febrile seizures.

Is there any contraindication of immunization for patients with metabolic disorders ?

- **All patients should receive age-appropriate immunizations in their respective schedules without delays.**
- **The only situation when vaccination may be contraindicated is with oral rotavirus vaccine in hereditary fructose intolerance patients with less than 9 kg due to amount of sucrose in the formulation (Vaccination strategies for people living with IEM in Brazil , December 2022)**
- **In Italy ; Safety of vaccines administration in HFI , 2020 :**

- **Rotarix pre-established oral suspension (clear and colourless liquid , the only formulation commercially available in Italy) and Rotateq contain sucrose above 1000mg / dose , therefore they are contraindicated in subjects with HFI.**
- **Rotarix (white powder and solvent for oral suspension , commercially available in other European Countries , containing sucrose 9 mg and sorbitol 13.5 mg can be administered in HFI children with a weight > 9.3 kg.**

Bear in mind:

Some vaccines contain a very small amount of phenylalanine.

The National Society for Phenylketonuria (NSPKU) advise the amount of phenylalanine contained in vaccines is negligible and therefore strongly advise individuals with phenylketonuria (PKU) to take up the offer of immunization. They are all safe to use in PKU.

Vaccination of people with primary and secondary immunodeficiencies (ACIP General Best Practice Guidelines for Immunization)

Vaccination of people with primary immunodeficiencies

Primary immunodeficiency	Specific immunodeficiency	Contraindicated Vaccines	Risk-specific recommended vaccines	Effectiveness and comments
B-lymphocyte (humoral)	Severe antibody deficiencies (e.g X-linked agammaglobulinemia & common variable immunodefiency	OPV , Smallpox ,LAIV, BCG, Ty21a (live typhoid), Yellow fever , MMR, MMRV	Pneumococcal Hib (Children 12-59 months of age)	The effectiveness of any vaccine is uncertain if it depends only on the humoral response (e.g,PPSV23) IGIV interferes with the immune response to measles vaccine & possibly varicella vaccine
	Less severe antibody deficiencies (e.g selective IgA deficiency & IgG subclass deficiency)	OPV , BCG, Yellow fever, Other live vaccines appear to be safe	Pneumococcal Hib (Children 12-59 months of age)	All vaccines likely effective; immune response might be attenuated
T-lymphocyte (cell-mediated & humoral)	Complete defects (e.g.,SCID disease , complete DiGeorge syndrome)	All live vaccines	Pneumococcal Hib (Children 12-59 months of age	Vaccines likely to be effective
	Partial defects (e.g.,most patients with DiGeorge syndrome, Wiskott-Aldrich syndrome, ataxia telangiectasia)	All live vaccines	Pneumococcal Meningococcal Hib (Children 12-59 months of age	Effectiveness of any vaccine depends on degree of immune suppression
	Interferon-gamma/Interleukin 12 axis deficiencies	All live bacterial vaccines (All live vaccines contraindicated in interferon-gamma or interferon-alpha deficiencies)	None	-
Complement	Persistent complement , properdin, or Factor B deficiency	None	Pneumococcal Meningococcal Hib (Children 12-59 months of age	All routine vaccines likely effective
	Taking eculizumab(Soliris) and/or ravulizumab(Ultomiris)	None	Meningococcal	-
Phagocytic Function	Chronic granulomatous disease	Live bacterial vaccines	None	Live viral vaccines likely safe and effective
	Phagocytic deficiencies that are undefined or accompanied by defects in T-cell & NK cell dysfunction (such as a Chediak-Higashi syndrome , Leukocyte Adhesion Deficiency[LAD], and myeloperoxidase deficiency)	MMR, MMRV, Varicella , OPV, Smallpox, BCG, LAIV , Ty21a , Yellow fever and bacterial vaccines	Pneumococcal	All inactivated vaccines safe and likely effective

Vaccination of people with secondary immunodeficiencies

Secondary immunodeficiency	Contraindicated vaccines	Risk-specific recommended vaccines	Effectiveness and comments
HIV/AIDS	OPV Smallpox BCG LAIV MMRV Withhold MMR & Varicella in severely immunocompromised people Yellow fever vaccine might have a contraindication or a precaution depending on clinical parameters of immune function	Pneumococcal Hib HepB MenACWY	MMR and Varicella in those with mild immunosuppression , rotavirus , and all inactivated vaccines including inactivated influenza as per routine vaccination schedule, might be effective
Generalized malignant neoplasm, transplantation, immunosuppressive or radiation therapy	Live viral and bacterial , depending on immune status	Pneumococcal Hib	Effectiveness of any vaccine depends on degree of immune suppression
Asplenia	LAIV	Pneumococcal Meningococcal Hib	All routine vaccines likely effective
Chronic renal disease	None	Pneumococcal HepB	All routine vaccines likely effective

Bear in mind :

- **Children 12-59 months : if unimmunized or received zero or only 1 dose , and that dose was administered before 12 months of age , should receive 2 Hib doses ,8 weeks apart ; if received 2 or more doses before age 12 months , and none after 12 months , should receive 1 Hib dose 8 weeks after the last dose ; if completed a primary series and received a booster dose at age 12 months or older , no additional Hib doses are recommended.**
- **There are no data to support IgA deficiency as a contraindication for yellow fever vaccine.**
- **Live bacterial vaccines : BCG and oral Ty21a Salmonella Typhi vaccine.**
- **Live viral vaccines : MMR , MMRV , OPV ,LAIV ,yellow fever , rotavirus , varicella , and vaccinia (smallpox). Nonemergency smallpox vaccination is not recommended for children younger than 18 years or the general public.**
- **Regarding T-lymphocyte immunodeficiency as a contraindication for rotavirus vaccine , data exist only for SCID.**
- **Symptomatic HIV infection or CD4+ T-lymphocyte count of < 200/mm3 or < 15 % of total lymphocytes for children aged < 6 years is a contraindication to yellow fever vaccine administration. Asymptomatic HIV infection or CD4+ T-lymphocyte count of 200-499/mm3 for persons aged ≥ 6 years or 15%-24% of total lymphocytes for children aged < 6 years is a precaution for yellow fever vaccine administration. Details of yellow fever vaccine administrations are available from CDC.**
- **HIV infected children should be considered for varicella vaccine if CD4 + T- lymphocyte count is ≥ 15% and should receive MMR vaccine if they are aged ≥ 12 months and do not have 1) evidence of current severe immunosuppression (i.e individuals aged ≤ 5 years must have CD4+T lymphocyte [CD4] percentages ≥ 15 % for ≥ 6 months ; and individuals aged > 5 years must have CD4+ percentages 15 % and CD4+ ≥ 200 Lymphocytes / mm3 for ≥ 6 months) and 2) other current evidence of measles , rubella , and mumps immunity. In cases when only CD4+ cell counts or only CD4+ percentages are available for those older than age 5 years, the assessment of severe immunosuppression can be based on CD4+ values (count or percentage) that are available. In cases when CD4+ percentages are not available for those aged ≤ 5 years, the assessment of severe immunosuppression can be based on age-specific CD4+ counts at the time CD4+ were measured ; i.e., absence of severe immunosuppression is defined as ≥ 6 months above age-specific CD4+ count criteria : CD4+ count > 750 lymphocytes /mm3 while aged ≤ 12 months and CD4+ count ≥ 500 lymphocytes / mm3 while aged 1 through 5 years.**
- **Withholding inactivated vaccines also is recommended with some forms of immunosuppressive therapy , like anti-CD20 antibodies , induction or consolidation chemotherapy, or patients with major antibody deficiencies receiving immunoglobulins. Inactivated influenza vaccine is an exception, but consideration should be given to repeating doses of any inactivated vaccine administered during these therapies.**

Concurrent or recent medication with vaccination :

Antibiotic therapy :

Antibiotic therapy doesn't interfere with response to non-live vaccines or most live vaccines with the following exceptions :

- **The Typho-O vaccine series should be finished at least 3 days before commencing , or initiated at least 3 days after completing , treatment with sulphonamides or other antibiotics active against S.typhi, or antimalarials. Exceptions include cholroquine , mefloquine and atovaquone/proguail (malarone), as these antimalarials do not affect the immune response to Typh-O vaccine and can be administered at the same time as , or at any interval before or after Typh-O vaccine.**
- **BCG vaccine should not be administered to individuals receiving drugs with anti-tuberculous activity, including fluoroquinolones.**

Antiviral therapy :

Antiviral therapy doesn't interfere with response to non-live vaccines or most live vaccines with the following exceptions :

- **Varicella-containing vaccines may have reduced effectiveness if given concurrently with antivirals active against varicella zoster virus (such as acyclovir , valacyclovir, famciclovir). People taking long-term antiviral therapy should discontinue these drugs , if possible , from at least 24 hours before administration of varicella-containing vaccine and should not restart antiviral therapy until 14 days after vaccination.**
- **LAIV should not be administered until 48 hours after antiviral agents active against influenza (e.g oseltamivir) are stopped, and antiviral agents shouldn't be administered until at least 14 days after receipt of LAIV unless medically indicated. If antiviral agents are administered within this time frame (from 48 hours before to 14 days after LAIV), revaccination should take place at least 48 hours after the antivirals are stopped.**

(Canadian Immunization Guide)

Vaccination of preterm infants :

➨ **Last recommendations from ACIP ;**

- **Regardless of birth weight , should be vaccinated at the same chronological age and according to the same schedule and using the same precautions as for full-term infants and children.**
- **Birth weight and size are not factors in deciding whether to vaccinate a clinically stable preterm infant except for hepatitis B vaccination.**
- **The full recommended dose of each vaccine should be used. Divided or reduced doses are not recommended.**

Revaccination in Pediatric Oncology Patients

- **In general, revaccination of a person after chemotherapy or radiation therapy is considered unnecessary if the previous vaccination occurred before therapy and not during therapy , with the exception of recipients of HSCT , who should be revaccinated. Determination of the level of immune memory and the need for revaccination should be made by the treating physician.(General Best Practice Guidelines for Immunization)**
- **Chemotherapy cause immunosuppression mostly during the induction and consolidation phases. After cessation of treatment, the immune system begins to recover within 3 months or more.**
- **There are no contraindications for inactivated or recombinant vaccines for these patients. However, live vaccines can cause vaccine-derived infections for a longer period after the chemotherapy in addition to the possible insufficient immune responses.**
- **OPV, LAIV, Oral typhoid, yellow fever and BCG vaccines are contraindicated for cancer patients after treatment.**
- **It's better to do MMR vaccination 6 months after completion of chemotherapy and varicella 1 year after chemotherapy. (The Eurasian Journal of Medicine)**

Recommendations For Vaccination In Patients With Cancer : IDSA

➨ **What Vaccines Should Be Given to Patients With Cancer ?**

- **Patients aged ≥ 6 months with hematological malignancies or solid tumor malignancies except those receiving anti-B-cell antibodies or intensive chemotherapy , such as for induction or consolidation chemotherapy for acute leukemia , should receive inactivated influenza vaccine (IIV) annually.**
- **PCV13 should be administered to newly diagnosed adults with hematological or solid malignancies and children with malignancies.PPSV23 should be administered to adults and children aged ≥ 2 years at least 8 weeks after the indicated dose(s) of PCV13.**
- **Inactivated vaccines (other than IIV) recommended for immunocompetent children in the CDC annual schedule can be considered for children who are receiving maintenance chemotherapy. However, vaccines administered during cancer chemotherapy shouldn't be considered valid doses unless there is documentation of a protective antibody level.**
- **Live viral vaccines should not be administered during chemotherapy.**
- **Three months after cancer chemotherapy , patients should be vaccinated with inactivated vaccines and the live vaccines for varicella , MMR according to the CDC annual schedule that is routinely indicated for immunocompetent persons. In regimens that included anti-B-cell antibodies , vaccinations should be delayed at least 6 months.**

Are there any contraindications for vaccinations in G6PD deficient patients?

- Considering that infection can precipitate acute haemolytic anemia ,which can in some case have severe outcomes , vaccination would confer a solid way to prevent this.
- Consulting the Vaccine Contraindications and Precautions provided by the Centers for Disease Control and Prevention (http://www.cdc.gov/vaccines/hcp/admin/contraindications.html) did not mention G6PD deficiency as proven contraindications for vaccination. Additionally, a search in Medline (a biomedical literature referencing more than 26 millions citations from journals and online books) does not seem to provide any studies or case report concerning the occurrence of adverse events in G6PD deficient patient after immunization either.
- One study advocated hepatitis A vaccination in patient with G6PD deficiency in order to prevent damaging consequences from intravascular hemolysis precipitated by acute hepatitis A infection in two children with G6PD deficiency.
- So far , there seems to be a lack of evidence against vaccination in G6PD patients. Like in many cases in medicine , the benefits of vaccination should be balanced with its possible adverse events.

Vaccine precautions in cardiac patients :

➡ **Precautions are recommended when vaccinating children with cardiac disease in the following circumstances :**

- **In children who are immunocompromised, live – attenuated vaccines may be contraindicated (refer to schedule , page : 166)**
- **children who have received blood products/and or immunoglobulin may need to delay vaccination (refer to schedule , page : 88)**
- **if children are also asplenic or have hyposplenism, further additional vaccines are recommended (refer to page : 156)**
- **Children for whom cardiac surgery is indicated :**

- **Before surgery – inactivated vaccines can be administered up to 1 week prior to surgery, live –attenuated vaccines can be administered up until 3 weeks prior to surgery (e.g. MMR , varicella)**
- **vaccines indicated following surgery should be delayed for at least one week due to the potential for confusing expected vaccine side effects with post-operative complications.**

- Vaccination are generally not recommended in the 3-4 weeks immaediately before or after a procedure or operations as they cause fever and general malaise which could either lead to the cancellation of the procedure or may wrongly alarm the child's medical team about post-operative infective complications.

Needle stick injuries in the community …

(Canadian Paediatric Society)

Management:

- After the injury, the wound should be cleaned thoroughly with soap and water as soon as possible. It should not be squeezed to induce bleeding.
- The extent of the wound, if any, or the probability of exposure of open skin lesions or mucous membranes to blood should be assessed.
- The child's immunization status for tetanus and HBV should be determined.
- Tetanus vaccine, with or without tetanus immunoglobulin, should be given if indicated.
- The circumstances of the injury should be documented (the date and time of injury or exposure where the needle was found , circumstances of the injury , type of needle , whether there was a syringe attached, whether visible blood was present in or on the needle or syringe , whether the injury caused bleeding and whether the previous user of the needle is known).
- Blood should be obtained from the child for:
- Baseline HBV, HIV and HCV status (may be stored for later testing).
- If antiretrovirals are being considered: complete blood count, differential, aspartate aminotransferase, alanine aminotransferase, alkaline phosphatase, blood urea nitrogen and creatinine.
- Testing needles and syringes for viruses is not indicated. Results are likely to be negative, result does not rule out possibility of infection.
- If the user of the needle is known, attempts should be made to assess for risk factors for blood-borne viruses and, if possible, to test for these viruses. Pending results, proceed as for an unknown source.

Interpretation of Hepatitis B Serologic Test Results

Hepatitis B serologic testing involves measurement of several hepatitis B virus (HBV)-specific antigens and antibodies. Different serologic "markers" or combinations of markers are used to identify different phases of HBV infection and to determine whether a patient has acute or chronic HBV infection, is immune to HBV as a result of prior infection or vaccination, or is susceptible to infection.

HBsAg **anti-HBc** **anti-HBs**	negative negative negative	Susceptible
HBsAg **anti-HBc** **anti-HBs**	negative positive positive	Immune due to natural infection
HBsAg **anti-HBc** **anti-HBs**	negative negative positive	Immune due to hepatitis B vaccination
HBsAg **anti-HBc** **IgM anti-HBc** **anti-HBs**	positive positive positive negative	Acutely infected
HBsAg **anti-HBc** **IgM anti-HBc** **anti-HBs**	positive positive negative negative	Chronically infected
HBsAg **anti-HBc** **anti-HBs**	negative positive negative	Interpretation unclear; four possibilities: 1. Resolved infection (most common) 2. False-positive anti-HBc, thus susceptible 3. "Low level" chronic infection 4. Resolving acute infection

Adapted from: A Comprehensive Immunization Strategy to Eliminate Transmission of Hepatitis B Virus Infection in the United States: Recommendations of the Advisory Committee on Immunization Practices. Part I: Immunization of Infants, Children, and Adolescents. MMWR 2005;54(No. RR-16).

DEPARTMENT OF HEALTH & HUMAN SERVICES
Centers for Disease Control and Prevention
Division of Viral Hepatitis

www.cdc.gov/hepatitis

- **Hepatitis B surface antigen (HBsAg):** A protein on the surface of hepatitis B virus; it can be detected in high levels in serum during acute or chronic hepatitis B virus infection. The presence of HBsAg indicates that the person is infectious. The body normally produces antibodies to HBsAg as part of the normal immune response to infection. HBsAg is the antigen used to make hepatitis B vaccine.
- **Hepatitis B surface antibody (anti-HBs):** The presence of anti-HBs is generally interpreted as indicating recovery and immunity from hepatitis B virus infection. Anti-HBs also develops in a person who has been successfully vaccinated against hepatitis B.
- **Total hepatitis B core antibody (anti-HBc):** Appears at the onset of symptoms in acute hepatitis B and persists for life. The presence of anti-HBc indicates previous or ongoing infection with hepatitis B virus in an undefined time frame.
- **IgM antibody to hepatitis B core antigen (IgM anti-HBc):** Positivity indicates recent infection with hepatitis B virus (≤6 mos). Its presence indicates acute infection.

- **The risk of acquiring HBV from an occupational needle stick injury when the source is hepatitis B surface antigen (HBsAg) positive ranges from 2% to 40%, depending on the source's level of viremia.**

- **The risk of acquisition of HIV from a hollow –bore needle with blood from a known HIV seropositive source is between 0.2 % and 0.5 %, based on prospective studies of occupational needle stick injuries. The risk is increased with higher viral inoculum, which is related to the amount of blood introduced and the concentration of virus in that blood.**

- **The risk of acquiring HCV as a result of an occupational needle stick injury when the source was infected varies from 3 % to 10 %.**

TABLE 1
Hepatitis B virus (HBV) prophylaxis

Child known to be HBV antibody- or HBsAg-positive	No action required
Child has not been fully vaccinated against HBV	Test for anti-HBs antibody and HBsAg. If results are not available in 48 h : - Give HBIG immediately (ideally within 48 h of injury ; efficacy unknown if > 7days after injury) Dose = 0.06 ml / kg intramuscularly - Give HBV vaccine (as soon as possible ,and at latest 7 days of injury) If anti-HBs antibody- and HBsAg-negative , complete vaccine series If anti-HBs- or HBsAg-positive , discontinue vaccine series. Arrange appropriate follow-up if HBsAg-positive.
Child has been fully vaccinated against HBV	Test for anti-HBs antibody. If results are not available in 48 h, give dose of HBV vaccine. If anti-HBs antibody-positive, no further action required. If anti-HBs antibody-negative, test for HBsAg: - If HBsAg-negative give HBIG and dose of HBV vaccine. - If HBsAg-positive , arrange appropriate follow-up.

HBIG HBV immunoglobulin;HBsAg Hepatitis B surface antigen

- **HIV Antiretroviral prophylaxis should be recommended only in cases of high risk , in which the source is considered likely to have HIV , the incident involved a needle and syringe with visible blood and blood may have been injected.**
- **In situations of low risk (source unlikely to have HIV , no visible blood in the device or superficial injury), prophylaxis should not be recommended but should be considered. Parents should be reassured of the low probability of their child acquiring HIV as a result of the incident.**
- **If the decision is made to begin antiretroviral prophylaxis :**

- **Antiretroviral should be started as soon as possible ideally within 1 to 4 h of the injury. Prophylaxis is not recommended if it can not be initiated within 72 h of the injury.**
- **If parents considering prophylaxis are undecided, they should be advised that is preferable to start prophylaxis immediately and then discontinue if they wish because starting later may be of no benefit.**
- **The antiretroviral agents used should be those currently recommended for occupational and nonoccupational exposures.**

• **for low-risk situations , zidovudine plus lamivudine.**
• **for high-risk situations , add lopinavir/ritonavir.**

- **The duration of prophylaxis is 28 days. For dosing and other details, refer to Table 2**

- **Antiretrovirals , especially protease inhibitors , may interfere with other medications. Check whether the child is taking other medications, and assess for possible interactions.**
- **Adverse effects : There are no data to suggest that a four- week course of antiretrovirals will have serious or long –term detrimental effects (listed in Table 2 footnote). Children with HIV infection have taken these drugs for years and serious side effects are rare.**
- **On the initial visit , drugs should be provided for two to three days and arrangements made for reassessment after that time to review adherence, assess adverse effects and arrange further follow-up. If the decision is made to continue prophylaxis, prescribe drugs to complete the 28-day course.**

TABLE 2

Antiretroviral agents recommended for HIV postexposure prophylaxis

Agent	Dosage	Comments
Nucleoside reverse transcriptase inhibitors* Zidovudine (ZDV)	6 weeks to 12 years : 160 mg/m2/dose tid or 240 mg/m2/dose bid ≥ 12 years : 300 mg/dose bid	Available in oral solution 10 mg/ml ;100 mg capsules; 300 mg tablets. Can be taken with or without food; may be better tolerated with food
Lamivudine (3TC)	1 month to 16 years : 4 mg/kg/dose bid (maximum 150 mg/dose) ≥ 16 years and ≥ 50 kg : 300 mg once daily	Available in oral solution 10 mg/ml ;150 mg capsules; 300 mg tablets. Can be taken with or without food; may be better tolerated with food
ZDV + 3TC (Combivir)**	13 years and > 37 kg : One tablet bid	Tablet contains 300 mg ZDV plus 150 mg 3TC
Protease inhibitor*** Lopinavir/ritonavir(LPV/RTV)	6 months to 12 years $: (230 mg LPV/57.5 RTV) m2/dose bid (maximum 400 mg LPV/100 mg RTV)/dose >12 years :(400 mg LPV/100 mg RTV)dose bid	Available as oral solution (80 mg LPV/20 mg RTV) ml ; 200 mg LPV/50 mg RTV tablets. Should be taken with a high-fat meal

**ZDV and 3TC are well tolerated. Occasionally children have anorexia , nausea , vomiting , diarrhea , abdominal pain , fatigue and headache. Asymptomatic mild neutropenia , anemia or elevation of liver enzymes may occur, which resolve after treatment is completed ; **GlaxoSmithKline Inc , Canada ; ***LPV/RTV may cause nausea , vomiting , diarrhea or abdominal discomfort; $Alternative dosing - < 15 kg :*
(12 mg/kg LPV/3 mg/kg RTV) dose twice a day (bid) ; 15-40 kg : (10 mg /kg LPV /2.5 mg / kg RTV)/dose bid. tid Three times a day

Unfortunately , there is no effective HCV postexposure prophylaxis at present.

- Alpha – interferon and ribavirin are used in therapy of chronic HCV infection , but their benefit for prophylaxis is not known.
- It is important to determine whether a potential exposure results in transmission of HCV because 50% to 60% of infected children will have persistent , asymptomatic infection for which follow-up by a specialist is indicated. Chronic hepatitis will eventually develop in some of these cases, and antiviral treatment may be required.

Follow-up

- Arrange follow-up and advise parents of the need for it (e.g, monitoring of side effects if on antiretroviral prophylaxis , testing for acquisition of infection and completion of HBV vaccination).
- If receiving antiretroviral prophylaxis :
- Reassess at two to three days , by phone or visit.
- Follow-up at two , four and six weeks for complete blood count , differential , aspartate aminotransferase, alanine aminotransferase , blood urea nitrogen and creatinine.
- At four weeks , give second HBV vaccine dose if only one previous dose received (consult table 1) or if no antibody or antigen detected on initial testing.
- At six weeks , test for anti-HIV antibody.
- At three months, test for anti-HIV antibody (unless previously positive) and anti-HCV antibody.
- At six months , test for anti-HIV , anti-HCV and anti-HBsAg antibody (unless previously positive). Give third HBV vaccine dose if only two previous doses received.
- If anti-HBs antibody negative at six months , test again one to two months after the third dose of vaccine. If still negative , test for HBsAg. If negative for both , give a fourth dose of HBV vaccine and test again one to two months later. If still negative refer to an appropriate specialist.
- If HIV , HCV or HBV infection occurs , test the stored baseline sera (unless already done) to determine whether infection was subsequent to the injury, and arrange for appropriate follow-up.

Meningitis – prophylaxis for contacts including Staff

NHS

1.Meningococcal Meningitis or Septicaemia

➡**Chemoprophylaxis :**

- **Should be offered to all individuals who have had *prolonged close contacts* with the case in a *household type setting* during the 7 days before the onset in the case.**
- **Staff only require chemoprophylaxis if they have administered mouth-to-mouth resuscitation or have been directly exposed to large particle droplets/secretions from the respiratory tract of a case.**
- **chemoprophylaxis should be given as soon as possible , ideally within 24 hours , after the diagnosis of the index case**

Table 1: Recommended chemoprophylaxis for Meningococcal meningitis or septicaemia [1, 2]:

1st line: (for all age groups and in pregnancy)	**Ciprofloxacin** Adults and children over 12 years: 500mg PO as a single dose Children 5 to 12 years: 250mg PO as a single dose Children < 5 years - 4kg or over: 125mg PO as a single dose - less than 4kg: 30mg/kg PO as a single dose Pregnancy – short duration of treatment with ciprofloxacin is considered safe.[1, 5] Breastfeeding – short duration of treatment with ciprofloxacin is considered safe.[1, 5] **Prescribers:** • Avoid if known ciprofloxacin hypersensitivity (see alternative below). • Refer to counseling points on the 'Patient Information Leaflet for Ciprofloxacin' (Appendix 1). • Prescribe using a separate outpatient prescription form for each individual. **Nurses:** • Obtain ciprofloxacin from ward stock. • Give under supervision and endorse administration on the outpatient prescription form. • Provide patient with 'Patient Information Leaflet for Ciprofloxacin' (see Appendix 1) and a copy of the outpatient prescription form.
Alternative: (if ciprofloxacin is contra-indicated e.g. known ciprofloxacin allergy)	**Rifampicin** Adults and children over 12 years: 600mg PO 12 hourly for 2 days Children - 1 to 12 year: 10mg/kg (max. 600mg) PO 12 hourly for 2 days - less than 1 year: 5mg/kg PO 12 hourly for 2 days ***If weight not known*** ***Doses based on average weight for age[1]*** 0 – 2 months: 20mg = 1ml x 100mg/5ml syrup BD for 2 days 3 – 11 months: 40mg = 2ml x 100mg/5ml syrup BD for 2 days 1 – 2 years: 100mg = 5ml x 100mg/5ml syrup BD for 2 days 3 – 4 years: 150mg = 7.5ml x 100mg/5ml syrup BD for 2 days 5 – 6 years: 200mg = 10ml x 100mg/5ml syrup BD for 2 days 7 – 12 years: 300mg (as capsule or syrup) BD for 2 days Pregnancy – NOT recommended.[1, 5] Breastfeeding – considered safe.[1, 5] **Prescribers:** • Avoid if jaundice, known rifampicin allergy or pregnant (seek pharmacist advice for alternatives such as Ceftriaxone IM stat or Azithromycin PO stat).[1] • Interaction with anticoagulants, antiepileptics, antiretrovirals and oral contraceptives – counsel patient and refer to GP or clinic as appropriate. • Refer to counseling points on the 'Patient Information Leaflet for Rifampicin' (Appendix 2). • Prescribe using a separate outpatient prescription form for each individual. **Nurses:** • Obtain supply of rifampicin from pharmacy or the on-call pharmacist if out-of-hours. • Endorse supply on the outpatient prescription form. • Provide patient with 'Patient Information Leaflet for Rifampicin' (see Appendix 1) and a copy of the outpatient prescription form.

1.Meningococcal Meningitis or Septicaemia

Immunization :

Immunoprophylaxis of Close Contacts ; identify those close contacts of confirmed cases of Group A, C, W135 or Y who are at highest risk of meningococcal disease and for whom immunization is indicated in addition to chemoprohylaxis.

Close Contacts Recommended to Receive Both Immunoprophylaxis (Immunization) and Chemoprophylaxis
• Household contact of the case • Persons who share sleeping arrangement with the case • Persons who have had direct contamination of their nose or mouth with oral/nasal secretions of a case (i.e kissing on the mouth ; sharing toothbrushes, eating utensils , cigarettes, mouth-guards, water bottles, or musical instrument mouthpieces) • Children and staff in child care and preschool facilities

Household contacts in particular have an increased risk of re-exposure to the bacteria that persists for up to one year after disease in the index case and beyond any protection from chemoprophylaxis. In general , this prolonged risk is not seen among other contacts that do not have ongoing exposure.

Assess the immunization status of close contacts who are recommended to receive immunoprophylaxis in addition to chemoprophylaxis. Ascertain whether meningococcal vaccine(s) has been received in the past including type of vaccine, number of doses and age or date at time of administration.

Chemoprophylaxis should be given first and the decision to offer vaccine should be made when the results of typing are available. Appropriate vaccine should be given within 4 weeks after illness onset.

Table 2: Recommended vaccination for index cases and contacts of Meningococcal meningitis or septicaemia: [1, 2]

Patient status	Vaccine schedule
Confirmed serogroup C infections	
INDEX CASE	
Not immunised with MenC vaccination AND under 25 years	Give **MenC** vaccination according to recommended UK schedule - see The Green Book
CLOSE CONTACT OR IMMEDIATE FAMILY OF CASES	
Not previously immunised with MenC vaccination	Give **MenC** vaccination according to recommended UK schedule - see The Green Book
Partially immunised with MenC vaccination	Complete the course of **MenC** vaccination according to UK schedule - see The Green Book
Immunised with MenC vaccination only in infancy	Give **MenC** vaccine: - single dose of 0.5ml IM
Completed the recommended immunisation course of MenC vaccination (including the 12-month booster) more than 1 year prior	

Patient status	Vaccine schedule
Confirmed serogroup A, W135 or Y infections	
INDEX CASE	
Not immunised with MenC vaccination AND under 25 years	Give **MenC** vaccination according to recommended UK schedule - see The Green Book
CLOSE CONTACT OR IMMEDIATE FAMILY OF CASES	
All close contacts	Give **MenACWY conjugate** vaccine: *Children < 1 year:* - two doses of 0.5ml IM given one month apart *Children ≥ 1 year and Adults:* - single dose of 0.5ml IM

Abbreviations: **MenC** = Meningitis C conjugate vaccine; **MenACWY** = quadrivalent ACWY conjugate vaccine

Refer to Public Health England - Immunisation against Infectious Disease: The Green Book for up-to-date information and changes to the immunisation schedule.

2.Haemophilus Influenza Disease (Meningitis, Septicaemia, Epiglottitis)

- Since the introduction of Hib vaccine , invasive *Haemophilus Influenza* type b (Hib) disease is now rare.

- **Chemoprophylaxis :**

Chemoprophylaxis should be given to all members of the household contacts *only when one or more family members are at risk.* At risk individuals include those who are immunocompromised, asplenic or children under 10 years of age.

Index case should be given immunoprophylaxis if they are under 10 years of age or if they are in contact with at risk household contacts. Not all children have been shown to mount an antibody response following an invasive Hib disease.

Table 3: Recommended chemoprophylaxis for *Haemophilus Influenzae* Disease [3, 4]:

1st line:	**Rifampicin** Adults and children > 12 years: 600mg PO once daily for **four** days Children > 3 months – 12 years: 20mg/kg (max. 600mg) PO once a day for **four** days Children 1 month – 3 months: 10mg/kg PO once a day for **four** days
	Pregnancy – NOT recommended. [1, 5]
	Breastfeeding – considered safe. [1, 5]
	Prescribers: • Avoid if jaundice, known rifampicin allergy or pregnant (see alternative below). • Interaction with anticoagulants, antiepileptics, antiretrovirals and oral contraceptives – counsel patient and refer to GP or clinic as appropriate. • Refer to counseling points on the 'Patient Information Leaflet for Rifampicin' (Appendix 2). • Prescribe using a separate outpatient prescription form for each individual. **Nurses:** • Obtain supply of rifampicin from pharmacy or contact on-call pharmacist if out-of-hours. • Endorse supply on the outpatient prescription form. • Provide patient with 'Patient Information Leaflet for Rifampicin' (see Appendix 1) and a copy of the outpatient prescription form.

Alternative: (if rifampicin is contra-indicated)	**Ceftriaxone** Adults and children > 12 years: 1g IM or IV once a day for 2 days Children 1 month – 12 years: 50mg/kg (max.1g) IV infusion once a day for 2 days
	Pregnancy – considered safe. [1, 5]
	Breastfeeding – considered safe. [1, 5]
	Prescribers: • Avoid if known beta-lactam allergy (seek Microbiology advice). • Interaction with anticoagulants – counsel patient and refer to GP or clinic as appropriate. • Prescribe using a separate outpatient prescription form for each individual. **Nurses:** • Obtain ceftriaxone from ward stock. For IV bolus (adult), reconstitute 1g vial with 10ml WFI and give over 2 – 4minutes. For IM (adult), reconstitute 1g vial with 3.5ml of lidocaine 1%. For IV infusion (children), reconstitute 1g vial with 10ml WFI (resultant solution of approx. 100mg/ml) then draw up the required dose and add to 50ml N/S. Infuse over 30 minutes. [4, 7] • Endorse administration on the outpatient prescription form. • Advise when the next dose is due and make arrangements to give the outstanding dose. • Provide patient with a copy of the completed outpatient prescription form.

2.Haemophilus Influenza Disease(Meningitis, Septicaemia , Epiglottitis)

- **Immunization :**

Any unimmunized index cases and household contacts from 2 months up to 10 years of age should be immunized as appropriate

The Public Health department should take the opportunity to immunize any unimmunized child contacts < 10 years of age ,e.g. at nursery , playgroup or school.

Table 4: Recommended vaccination for index cases and contact of *Haemophilus Influenzae* Disease: [3, 4]

Patient's status	Vaccine
INDEX CASE – 2 month to ≤ 10 years of age	
Not immunised Partially immunised	Should complete their age-specific course of immunisation after recovery from infection.
Previously vaccinated against Hib	Should have convalescent antibody levels measured and booster vaccination may be advised. *If antibody testing is not possible:* An additional dose of Hib-containing vaccine should be given after recovery from infection.
HOUSEHOLD CONTACTS – 2 month to ≤ 10 years of age	
Never received any immunisation	**DTaP/IPV/Hib vaccine (Pediacel®)** - 3 doses of 0.5ml IM given one month apart.
Immunisation interrupted or incomplete	Complete the UK recommended immunisation schedule - this should be resumed but not repeated.
If never received Hib-containing vaccine BUT completed primary course of diphtheria, tetanus, pertussis and polio	*Children < 1 year:* **Hib / MenC vaccine (Menitorix®)** - 3 doses of 0.5ml IM given one month apart. *Children 1 to 10 years:* **Hib / MenC vaccine (Menitorix®)** - single dose of 0.5ml IM.
Received Hib vaccine in infancy BUT did not receive a booster dose of Hib-containing vaccine after the age of 12 months	**Hib / MenC vaccine (Menitorix®)** - single dose of 0.5ml IM.

Refer to Public Health England - Immunisation against Infectious Disease: The Green Book for up-to-date information and changes to the immunisation schedule.

3.Pneumococcal Disease

➡ **No Chemoprophylaxis is required.**

Close contacts of pneumococcal meningitis or other invasive pneumococcal disease are not normally at increased risk of pneumococcal infection and therefore antibiotic prophylaxis is not indicated.

Any case of invasive pneumococcal infection or lobar pneumonia believed to be due to *Streptococcus pneumoniae* should prompt a review of the patient's medical history to establish whether they are in a recognized risk group and whether they have been vaccinated.

Should infants exposed to biologics be immunized? (Mothers receiving biologics during pregnancy) (Canadian Medical Association Journal | CMAJ)

All exposed infants should receive inactivated immunizations according to the routine schedule. A few small studies have evaluated the immunogenicity of vaccines and have reported normal and protective antibody titers in exposed infants , similar to nonexposed infants. Two studies showed lower antibody levels to *Hemophilus influenza* type B (Hib) after Hib-conjugate vaccination in exposed infants compared with nonexposed infants.

Most guidelines recommend avoiding all live vaccines for the first 6-12 months of life. A single case of disseminated bacille Calmette-Guerin (BCG) disease was reported in an infant exposed to infliximab following the live attenuated BCG vaccine. However , cohort studies of infants receiving BCG have reported no serious adverse outcomes.

In Canada , the only live vaccine that is routinely administered before 6 months of age is the rotavirus vaccine. Accumulating clinical experience suggests that the rotavirus Vaccine may be given safely to certain infants exposed to biologics, even if the drug is still detectable in serum. Case series and more recently , cohort studies have described exposed infants who received this vaccine without serious complications, such as vaccine-associated rotavirus disease. Rotavirus infection from the live-attenuated vaccine has been limited primarily to patients with severe combined immune deficiency, suggesting that this adverse event is mostly observed in children with severe T- and B- cell immunodeficiency and not with other immune defects or mild immunosuppression.

Specialist assessment of immune function is recommended before considering administration of rotavirus vaccination, with careful review of the specific drug exposure. This information should then guide a risk-benefit discussion about whether or not to proceed with this vaccine.

If the exposed infant cannot be evaluated , then live vaccines should be avoided for the first 6-12 months of life. In certain situations (e.g., travel or local outbreak), the theoretical risk of providing live vaccines before 12 months of age should be weighed against the risks of exposure to natural infection.

Live vaccines are generally permitted after 12 months of age, when all types of biologics would be cleared from the infant's circulation.

Table 1: Characteristics of monoclonal antibody biologics and indications for use

Biologic type	Drug name	Structure	Indication for use*
Anti-TNFα	Infliximab	Chimeric anti-TNFα IgG1	Rheumatoid arthritis, ankylosing spondylitis, Crohn disease, ulcerative colitis, plaque psoriasis
	Adalimumab	Recombinant humanized anti-TNFα IgG1	Rheumatoid arthritis, polyarticular JIA, psoriatic arthritis, ankylosing spondylitis, Crohn disease, ulcerative colitis, hidradenitis suppurativa, plaque psoriasis, uveitis
	Golimumab	Humanized anti-TNFα IgG1	Rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, nonradiographic axial spondyloarthritis, ulcerative colitis
	Certolizumab pegol	Recombinant, humanized antibody to the antigen-binding fragment to anti-TNFα	Rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, plaque psoriasis
	Etanercept	Human recombinant TNFα receptor/IgG1-Fc fusion protein	Ankylosing spondylitis, rheumatoid arthritis, polyarticular JIA, psoriatic arthritis, plaque psoriasis
Anticytokine	Ustekinumab	Anti-IL-12 and IL-23 humanized IgG1	Plaque psoriasis, psoriatic arthritis, Crohn disease, ulcerative colitis
	Tocilizumab	Anti-IL-6 receptor humanized IgG1	Rheumatoid arthritis, polyarticular and systemic JIA, giant cell arteritis
	Canakinumab	Anti-IL-1β human IgG1	CAPS, TRAPS, HIDS, MKD, FMF, systemic JIA
Anti-integrin	Vedolizumab	Humanized anti- α4β7 integrin IgG1	Ulcerative colitis, Crohn disease
	Natalizumab	Anti-integrin α4 subunit humanized IgG4	Multiple sclerosis
Anti-B cell	Rituximab	Anti-CD20 IgG1	Rheumatoid arthritis, non-Hodgkin lymphoma, chronic lymphocytic leukemia, granulomatosis with polyangiitis
	Belimumab	Anti-B-cell activating factor human IgG1	Systemic lupus erythematosus

Note: CAPS = cryopyrin-associated periodic syndromes, FMF = familial Mediterranean fever, HIDS = hyperimmunoglobulin D syndrome, Ig = immunoglobulin, IL = interleukin, JIA = juvenile idiopathic arthritis, MKD = mevalonate kinase deficiency, TNF = tumour necrosis factor, TRAPS = tumour necrosis factor receptor associated periodic syndrome.
*Off-label use not listed.

Table 2: Drug transfer, estimated drug clearance and clinical experience of monoclonal antibody biologics in pregnancy

Biologic	Drug transfer to fetus	Estimated drug clearance in the infant	Level of clinical experience*	Reference
Infliximab	High	3–7 mo	++++	2,15,33,36,44
Adalimumab	Moderate	3–5 mo	++++	2,15,33,44
Golimumab	Moderate	Unknown	+	38
Certolizumab pegol	Minimal (passive diffusion)	NA	+++	2,15,41,42,45-47
Etanercept	Low	0–3 mo	+++	39,40
Ustekinumab	Moderate	Unknown	+	15,37,38
Vedolizumab	Low-moderate	Likely < 3 mo	+	15,38,43,44
Natalizumab	Low-moderate	Unknown	+	15,38,48,49
Rituximab	Moderate-high	Unknown	+	50,51
Belimumab	Unknown	Unknown	+	52

Note: NA = not applicable.

*We categorized the amount of clinical experience into 4 levels depending on size and quality of studies: + = least clinical experience (i.e., small case reports and case series only) to ++++ = largest clinical experience (i.e., large prospective cohort studies with > 1000 participants enrolled).

At the end ,

I hope I have completed what you want to know about these topics. Talking about that was not easy. I did my best to you to enjoy reading.

Thank you